National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Prostate Cancer Early Detection

Version 2.2026 — February 18, 2026

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NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

NCCN Guidelines Index
Table of Contents
Discussion

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National Comprehensive Cancer Network®

NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

NCCN Guidelines Index
Table of Contents
Discussion

NCCN Prostate Cancer Early Detection Panel Members
Summary of the Guidelines Updates
Introduction (PROSD-1)
Baseline Evaluation, Risk Assessment, and Early Detection Evaluation (PROSD-2)
Further Evaluation and Indications for Biopsy (PROSD-3)
Management of Biopsy Results (PROSD-4)
Principles of Biomarker Testing in Prostate Cancer Early Detection (PROSD-A)
Abbreviations (ABBR-1)

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See NCCN Categories of Evidence and Consensus.

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National Comprehensive Cancer Network®

NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

NCCN Guidelines Index
Table of Contents
Discussion

Updates in Version 2.2026 of the NCCN Guidelines for Prostate Cancer Early Detection include:

MS-1

- The Discussion section has been updated to reflect the changes in the algorithm.

Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer Early Detection from Version 2.2025 include:

Global

- Modifications made throughout the Guidelines related to terminology for biomarker testing and results.

PROSD-1

- Footnote a added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).

PROSD-2

- Risk Assessment, sentence removed: Discuss potential benefits and harms of prostate cancer early detection.
- Top pathway, column 3, header modified: Age 40–75 y for ~~patients with high-risk~~:
- Middle pathway, column 3, header modified: Age 45–75 y for *all others* ~~patients with average risk~~:

PROSD-3

- Further Evaluation and Indications for Biopsy
  - Column 2
    - ◊ Bullet 1 modified: Multiparametric MRI (mpMRI) (category 1)*, including PSA density* ~~if available.~~
    - ◊ Bullet 2 modified: Consider biomarkers *(category 2B)*...
- Footnote n modified: A negative MRI does not exclude the possibility of cancer. Consider biomarkers, risk calculators, and/or PSA density when deciding whether to avoid a biopsy in an individual with a negative mpMRI result. Biparametric MRI *and micro-ultrasound (at an experienced center)* have been shown to offer similar diagnostic quality and can be considered as an alternative to mpMRI in certain contexts. Bosaily AE, et al. Eur Urol 2020;78:503-511; van der Leest M, et al. Eur Urol 2019;76:574-581; Schoots IG, et al. AJR Am J Roentgenol 2021;216:3-19; *Kinnaird A, et al. JAMA 2025;333:1679-1687.*
- Footnote o modified: Lower percent-free PSA and/or higher PSA density are associated with a greater risk of high-grade prostate cancer. *Persistent acceleration in PSA over and above an established baseline is also concerning for high-risk prostate cancer. Patients with a persistent and significant increase in PSA should be encouraged to undergo biopsy, including patients with a higher PSA density, even if they have a normal MRI. Haj-Mirzaian A, et al. JAMA Netw Open 2024;7:e244258.*
- Footnote s added: Micro-ultrasound–guided biopsy performed at expert centers was found to be non-inferior to MRI-targeted fusion biopsy for the detection of clinically significant prostate cancer. Kinnaird A, et al. JAMA 2025;333:1679-1687.
- Footnote removed: Patients with a persistent and significant increase in PSA should be encouraged to undergo biopsy, including patients with a higher PSA density, even if they have a normal MRI. Haj-Mirzaian A, et al. JAMA Netw Open 2024;7:e244258.

PROSD-4

- Management of Biopsy results, bottom pathway modified
  - Bullet 1
    - ◊ Sub-bullet 1 added: Obtain mpMRI

Continued
UPDATES

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Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer Early Detection from Version 2.2025 include:

◊ Sub-bullet 2 modified: Consider biomarkers that improve the specificity of screening *(category 2B)* ~~and/or mpMRI~~.

▸ Bullet 2, sub-bullet 2 modified: Consider biomarkers that improve the specificity of screening *(category 2B).*

- Footnote v, last sentence modified: It is well known that a negative prostate biopsy does not preclude a diagnosis of prostate cancer on subsequent biopsy. Patients with negative prostate biopsies should be followed with DRE and PSA with ~~consideration of~~ mpMRI *(if no prior high-quality mpMRI)* and *consideration of* biomarker tests that improve the specificity of PSA testing.

PROSD-A

- Bullet 1 modified: ~~Biomarkers that improve the specificity of detection are not, as yet, mandated as first-line screening tests in conjunction with serum PSA. However, there may be some p~~Patients who meet PSA standards for consideration of prostate biopsy *should undergo mpMRI (category 1). If mpMRI is not available, contraindicated, or of low-quality or if~~, ~~but for whom~~ the patient and/or the physician wish to more precisely estimate risk *of high-grade cancer (Grade Group ≥2) prostate cancer compared to PSA, certain biomarker tests can be considered before proceeding to immediate biopsy (category 2B).*
- Bullet 2 added: Patients with a negative biopsy can also consider the use of biomarkers (category 2B) if there remains suspicion that clinically significant prostate cancer may be present and if a high-quality mpMRI has already been obtained.
- Bullet 3 added: RISKS: Clinicians and patients should be aware that the extent of validation of these tests across diverse populations is variable and is mostly limited to retrospective observational studies, often with industry support (see Discussion). There are risks in using biomarker tests that lack robust validation to change management decisions, as they may drive patients or providers to inappropriate options. In particular, these tests may underestimate risk, leading to delayed or missed diagnoses.
- Bullet 4 added: The Panel emphasizes that these tests should not be used in place of mpMRI.
- Bullet 5, sub-bullet 5, sub-sub-bullet 2, sub-sub-sub-bullet 1 modified: Reduction in clinically insignificant cancers, >~~50~~*30%.*
- Bullet 6 modified: Prior to initial biopsy: ~~The probability of high-grade cancer (Grade Group ≥2) may be further defined prior to initial biopsy utilizing t~~Tools *that meet most or all of the above criteria based on available data include* ~~such as~~ the Prostate Health Index (PHI), SelectMDx, 4Kscore, ExoDx Prostate Test, MyProstateScore (MPS), MPS 2.0 (MPS2) ~~(category 2B)~~, Stockholm3, and IsoPSA.
- Bullet 7 modified: After initial *negative* biopsy: *Tools that meet most or all of the above criteria based on available data include* ~~Tests that improve specificity in the post-biopsy setting may be considered in patients thought to be higher risk despite a negative prostate biopsy. Tests to be considered are~~ percent-free PSA, 4Kscore, PHI, ~~PCA3,~~ ConfirmMDx, ExoDx Prostate Test, MPS, MPS2 ~~(category 2B)~~, and IsoPSA.
- Bullet 8 added: The Panel encourages researchers to perform randomized controlled trials that validate the safety and efficacy of biomarker tests; that compare biomarker tests to mpMRI; and that compare biomarker tests head-to-head.
- Bullet removed: The extent of validation of these tests across diverse populations is variable.
- Bullet removed: It is not yet known how such tests could be applied in optimal combination with MRI.

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NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

NCCN Guidelines Index
Table of Contents
Discussion

INTRODUCTION[a]

The Panel recognizes that prostate cancer represents a true spectrum of disease and that *not* all patients diagnosed with prostate cancer require treatment. The Panel believes that maximizing the detection of early prostate cancer will increase the detection of both indolent (slower-growing) and aggressive (faster-growing) prostate cancers. The challenge is to minimize immediate treatment (overtreatment) of indolent cancers by accurately characterizing the biology of the detected cancer. This guideline highlights several techniques designed to improve the identification of significant cancer while avoiding the detection of indolent disease. Identification and selective treatment of aggressive cancers should result in significant decreases in morbidity and mortality while limiting adverse effects on quality of life. The NCCN Guidelines for Prostate Cancer Early Detection do not address the treatment of prostate cancer. See the NCCN Guidelines for Prostate Cancer for prostate cancer treatment recommendations. It is the intention of the Panel that these guidelines be linked. Specifically, early detection strategies that do not recognize the importance of refined and selective treatment may result in harm.

The guidelines are specifically for individuals with a prostate opting to participate in an early detection program (after receiving the appropriate counseling on the potential benefits and harms). It is the majority opinion of the Prostate Cancer Early Detection Panel members that there is a growing population of patients currently being diagnosed with prostate cancer who can, and should, be monitored for their disease rather than immediately treated as presented in the NCCN Guidelines for Prostate Cancer. The guidelines are continuously in a state of evolution, and the Panel will incorporate changes based on new evidence and expert opinion and provide a rating of consensus for each recommendation.

Baseline Evaluation (PROSD-2)

[a] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).

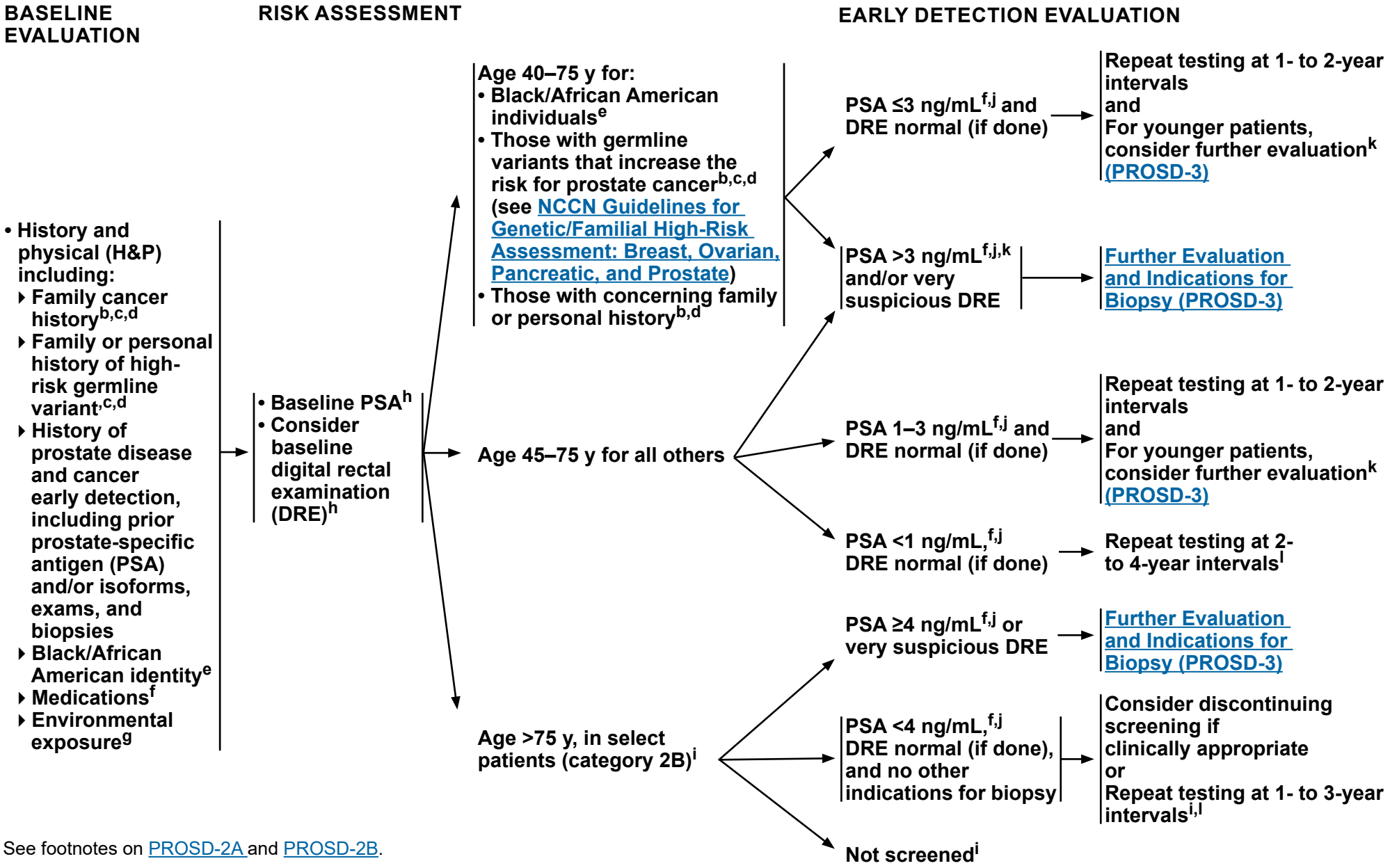
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NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

NCCN Guidelines Index
Table of Contents
Discussion



See footnotes on PROSD-2A and PROSD-2B.

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FOOTNOTES

[b] Family or personal cancer history and/or family or personal history of high-risk germline variants can inform when to begin shared decision-making regarding prostate cancer early detection. Patients with a suspicious family history should undergo genetic testing. See NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate (CRIT-1) and NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric (LS-1). Page EC, Eur Urol 2019;76:831-842; Giri VN, et al. J Clin Oncol 2020;38:2798-2811.

[c] If there is a known or suspected cancer susceptibility gene, referral to a cancer genetics professional is recommended. Individuals harboring germline variants in prostate cancer risk genes may have an elevated lifetime risk of prostate cancer and, in the case of certain genetic variants, an elevated risk of early-onset and/or potentially lethal prostate cancer (eg, *BRCA2*). For prostate cancer susceptibility genes and cancer risk management based on genetic testing results, see NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate (GENE-A and BRCA-A).

[d] If genetic testing does not show the presence of a germline variant in a prostate cancer risk gene but family history is concerning, shared decision-making is recommended regarding the timing and frequency of PSA testing.

[e] Black/African American individuals have a significantly higher incidence of prostate cancer, earlier age at diagnosis, and increased prostate cancer mortality compared to white individuals. The higher incidence and earlier age at diagnosis are attributable to a greater incidence of preclinical prostate cancer and a higher likelihood that a preclinical tumor will spread. The survival disparities seen in Black/African American individuals are related to these factors and to social determinants of health, lack of access to care, and treatment disparities, and not to race itself or differences in the frequency of tumor-acquired genomic variants (Vince RA, et al. JAMA Netw Open 2023;6:e2250416; Schumacher FR, et al. JCO Precis Oncol 2021;5:PO.21.00324). In fact, data show that Black patients who receive equitable prostate cancer treatment have similar outcomes to white patients (Riviere P, et al. Cancer 2020;126:1683-1690). Therefore, there is a need for higher vigilance in this population in the form of appropriate screening and treatment. Although there is currently little evidence to support that screening at an earlier age will result in decreased morbidity and mortality compared to testing at age 45, and earlier screening may increase overdiagnosis (Tsodikov A, et al. Cancer 2017;123:2312-2319), data show that Black individuals benefit more from screening than white individuals (Basourakos SP, et al. NEJM Evid 2022;1:10.1056/evidoa2200031). Therefore, many support the recommendation for Black/African American individuals to consider beginning shared decision-making about PSA screening at age 40 years and to consider screening at annual intervals rather than every other year.

[f] Medications such as 5α-reductase inhibitors (finasteride and dutasteride) are known to decrease PSA by approximately 50%. PSA values in patients taking these medications should be corrected accordingly.

[g] Patients with exposure to Agent Orange (eg, many Vietnam War veterans) should be informed that Agent Orange exposure may be associated with an increased risk of high-grade prostate cancer.

[h] The best evidence supports the use of serum PSA for the early detection of prostate cancer. DRE should not be used as a stand-alone test. DRE can be considered as a baseline test in addition to serum PSA in all patients, but has its greatest usefulness in those with elevated PSA. Consider referral for biopsy or further testing if DRE is suspicious for cancer at any PSA level. Halpern JA, et al. J Urol 2018;199:947-953; Arsov C, et al. Int J Cancer 2022;150:1861-1869; Matsukawa A, et al. Eur Urol Oncol 2024;7:697-704.

Footnotes continued on PROSD-2B

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FOOTNOTES

[i] Prostate cancer early detection is recommended in adults with a prostate starting at age 45 years (or 40 years if risk factors, see risk assessment on PROSD-2) through 75 years who might have a life expectancy of ≥10 years and who opt to participate in an early detection program. The decision to test beyond 75 years should be individualized and include a discussion of the risks and benefits based on comorbidity status and estimated life expectancy. Testing after 75 years of age should be done only in very healthy people with little or no comorbidity (especially if they have never undergone PSA testing or have a rising PSA) to detect the small number of aggressive cancers that pose a significant risk if left undetected until signs or symptoms develop. Widespread testing in this population would substantially increase rates of overdetection and is not recommended.

[j] Transgender women and other gender-diverse individuals assigned male at birth retain their prostates regardless of gender-affirmation surgery. They should therefore undergo shared decision-making regarding age-appropriate prostate cancer early detection. For those on gender-affirming hormones, such as estrogen, PSA values should be interpreted cautiously bearing in mind the castrating effects of gender-affirming hormones that may drastically decrease PSA levels. One study found a median PSA level of 0.02 ng/mL (interquartile range [IQR] 0–0.2, 95th percentile 0.6 ng/mL) for transgender women with a median age of 60 years who were on estrogen for at least 6 consecutive months and had no known prostate cancer (Nik-Ahd F, et al. JAMA 2024;332:335-337). Clinicians should take this information into account when interpreting PSA values in transgender and other gender-diverse patients undergoing gender-affirming care. Special attention should be paid to rising PSA levels in this population.

[k] The median PSA values for those aged 40–49 years range from 0.5–0.7 ng/mL, and the 75th percentile values range from 0.7–0.9 ng/mL. Individuals who have a PSA above the median for their age group are at a higher risk for prostate cancer and aggressive prostate cancer. The higher above the median, the greater the risk. Further evaluation (PROSD-3) can be considered for patients who have PSA levels that are above the median for their age group, even if they do not meet the threshold of >3 ng/mL.

[l] Individuals ≥60 years with a PSA <1.0 ng/mL and those >75 years of age with a PSA <3.0 ng/mL have a very low risk of prostate cancer metastases in their lifetime. This low risk is especially true for those in the latter category. The testing interval ranges on PROSD-2 were not a unanimous decision by the Panel. There are some Panel members who believe that 2–4 years is too long for individuals aged 40–75 years with PSA >1.0 ng/mL. However the Panel majority supported the intervals as listed.

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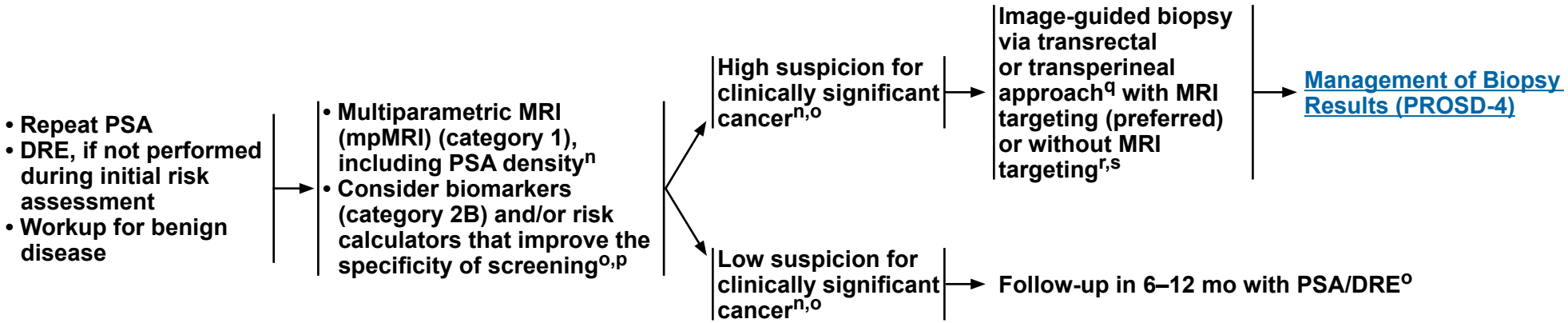
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NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

NCCN Guidelines Index
Table of Contents
Discussion

FURTHER EVALUATION AND INDICATIONS FOR BIOPSY[m]

MANAGEMENT



[m] The level of PSA correlates with the risk of prostate cancer. The Prostate Cancer Prevention Trial (PCPT) demonstrated that 15% of individuals with a PSA level ≤4.0 ng/mL and a normal DRE had prostate cancer diagnosed on end-of-study biopsies. Approximately 30% to 35% of those with serum PSA between 4 to 10 ng/mL will be found to have cancer. Total PSA levels >10 ng/mL confer a >67% likelihood of prostate cancer.

[n] A negative MRI does not exclude the possibility of cancer. Consider biomarkers, risk calculators, and/or PSA density when deciding whether to avoid a biopsy in an individual with a negative mpMRI result. Biparametric MRI and micro-ultrasound (at an experienced center) have been shown to offer similar diagnostic quality and can be considered as an alternative to mpMRI in certain contexts. Bosaily AE, et al. Eur Urol 2020;78:503-511; van der Leest M, et al. Eur Urol 2019;76:574-581; Schoots IG, et al. AJR Am J Roentgenol 2021;216:3-19; Kinnaird A, et al. JAMA 2025;333:1679-1687.

[o] Lower percent-free PSA and/or higher PSA density are associated with a greater risk of high-grade prostate cancer. Persistent acceleration in PSA over and above an established baseline is also concerning for high-risk prostate cancer. Patients with a persistent and significant increase in PSA should be encouraged to undergo biopsy, including patients with a higher PSA density, even if they have a normal MRI. Haj-Mirzaian A, et al. JAMA Netw Open 2024;7:e244258.

[p] Principles of Biomarker Testing in Prostate Cancer Early Detection (PROSD-A).

[q] A transperineal approach to transrectal ultrasound (TRUS) biopsy may be associated with a lower risk of sepsis and a reduced need for antibiotics compared to a transrectal approach.

[r] In patients undergoing biopsy, targeting using MRI/ultrasound fusion significantly increases the detection of clinically significant, higher-risk (Grade Group ≥3) disease while lowering the detection of lower-risk (Grade Group 1 or lower-volume Grade Group 2) disease. It is strongly recommended that image-guided biopsy techniques be employed routinely. Radiologic expertise and the use of high-quality mpMRI hardware is essential for optimal interpretation of scans. Most advocate for a combined targeted and systematic biopsy approach as some high-grade cancers are uniquely detected using the systematic approach and systematic biopsies are needed for risk stratification if cancer is found. However, some advocate for excluding systematic biopsy in those undergoing MRI targeting due to concerns that it may increase the risk of overdiagnosis. Siddiqui MM, et al. JAMA 2015;313:390-397; Ahmed HU, et al. Lancet 2017;389:815-822; Kasivisvanathan V, et al. N Engl J Med 2018;378;1767-1777; Drost FH, et al. Eur Urol 2020;77:78-94; Ahdoot M, et al. N Engl J Med 2020;382:917-928; Rouvière O, et al. Lancet Oncol 2019;20:100-109; van der Leest M, et al. Eur Urol 2019;75:570-578.

[s] Micro-ultrasound–guided biopsy performed at expert centers was found to be non-inferior to MRI-targeted fusion biopsy for the detection of clinically significant prostate cancer. Kinnaird A, et al. JAMA 2025;333:1679-1687.

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MANAGEMENT OF BIOPSY RESULTS

Cancer → **NCCN Guidelines for Prostate Cancer**

Intraductal carcinoma (IDC) without invasive carcinoma[t] → **NCCN Guidelines for Prostate Cancer**

Atypical intraductal proliferation (AIP) without invasive carcinoma[u] → **Repeat biopsy using MRI targeting and systematic biopsy to look for invasive carcinoma**

Atypia, suspicious for cancer →
High-grade prostatic intraepithelial neoplasia (PIN)[v,w] →
Benign[v,w,x] →

- **If no prior high-quality mpMRI:**
  - **Obtain mpMRI[v]**
  - **Consider biomarkers that improve the specificity of screening[p] (category 2B)**
  - **For atypia, consider repeated biopsy in 12–24 months with relative increased sampling of the atypical site**
- **If prior high-quality mpMRI:**
  - **PSA and DRE at 12- to 24-month intervals**
    and
  - **Consider biomarkers that improve the specificity of screening[p] (category 2B)**
- **Repeat prostate biopsy with refined biopsy techniques, if high suspicion of cancer[x]**

[p] Principles of Biomarker Testing in Prostate Cancer Early Detection (PROSD-A).

[t] IDC represents an independent adverse pathologic factor in both radical prostatectomy and needle biopsy specimens. Its presence in biopsy material strongly suggests the presence of high-grade cancer. Consideration should be given to initial treatment; otherwise, careful evaluation is indicated, with strong consideration given to repeat biopsy using MRI guidance. Porter LH, et al. Eur Urol 2017;72:492-495.

[u] Intraductal proliferations may show a greater degree of architectural complexity and/or cytologic atypia than typical high-grade PIN, yet falling short of the strict diagnostic threshold for IDC. The preferred terminology for these lesions is AIP. When diagnosed on needle biopsy, AIP is potentially considered a marker of unsampled cancer, and it is associated with an increased risk (50%) of invasive carcinoma and/or IDC on repeat biopsy. Shah RB, et al. Histopathology 2019;75:346-353.

[v] It is well known that a negative prostate biopsy does not preclude a diagnosis of prostate cancer on subsequent biopsy. Patients with negative prostate biopsies should be followed with DRE and PSA with mpMRI (if no prior high-quality mpMRI) and consideration of biomarker tests that improve the specificity of PSA testing.

[w] PSA testing may be discontinued at certain ages and PSA cutpoints. See Discussion.

[x] mpMRI and/or use of refined prostate biopsy techniques (image guidance using MRI/ultrasound fusion) may help identify regions of cancer missed on prior prostate biopsies and are recommended after at least 1 negative prostate biopsy and high suspicion for cancer based on PSA and/or biomarkers. mpMRI followed by lesion targeting increases the detection of clinically significant, higher-risk disease while lowering the detection of lower-risk disease. Although some advocate for excluding systematic biopsy in those undergo MRI targeting, most advocate for a combined approach as some high-grade cancers are uniquely detected using the systematic approach.

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PRINCIPLES OF BIOMARKER TESTING IN PROSTATE CANCER EARLY DETECTION

- Patients who meet PSA standards for consideration of prostate biopsy should undergo mpMRI (category 1). If mpMRI is not available, contraindicated, or of low-quality, or if the patient and/or the physician wish to more precisely estimate risk of high-grade (Grade Group ≥2) prostate cancer compared to PSA, certain biomarker tests can be considered before proceeding to immediate biopsy (category 2B).
- Patients with a negative biopsy can also consider the use of biomarkers (category 2B) if there remains suspicion that clinically significant prostate cancer may be present and if a high-quality mpMRI has already been obtained.
- RISKS: Clinicians and patients should be aware that the extent of validation of these tests across diverse populations is variable and is mostly limited to retrospective observational studies, often with industry support (see Discussion). There are risks in using biomarker tests that lack robust validation to change management decisions, as they may drive patients or providers to inappropriate options. In particular, these tests may underestimate risk, leading to delayed or missed diagnoses.
- The Panel emphasizes that these tests should not be used in place of mpMRI.
- Biomarker tests should meet the following criteria of both validity and clinical utility to be included in the NCCN Guidelines for Prostate Cancer Early Detection:
  - Test should be validated in a patient cohort that is representative of the intended-use population, ideally in a multi-site, prospective, external validation study or multiple single-site studies.
  - Studies should clearly and transparently describe the eligible patient population and how patients were selected for the study.
  - Validation study or studies should be appropriately powered to achieve statistical significance (on the order of hundreds of patients or more, rather than tens of patients).
  - Study populations should ideally reflect the racial and ethnic diversity of the target population, and studies should report racial and ethnic composition.
  - Validation studies should report most, or all, of the measures listed below (acceptable thresholds are underlined). Reports of receiver operating characteristic (ROC) and area under the curve (AUC) measures alone are not adequate, because these measures are difficult to interpret and they do not describe performance in relation to any specific clinical cut-off.
    - ◊ Number of patients with clinically significant prostate cancer (currently defined as Grade Group ≥2) diagnosed with the test above a specific test cut-off compared to without the test (PSA alone) at specific test cutoff
      - – Maintained detection (non-inferior to PSA)
      - – Missed/delayed clinically significant diagnoses, <10%
    - ◊ Number of patients with clinically insignificant prostate cancer (Grade Group 1) found with the test and without the test (PSA alone) above and at a specific test cutoff
      - – Reduction in clinically insignificant cancers, >30%
      - – Ratio of reduction in clinically insignificant to missed/delayed clinically significant diagnoses, ≥2
    - ◊ Number of biopsies performed in individuals with the test above a specific test cut-off compared to without test (PSA alone)
      - – Unnecessary biopsies avoided, >30%
- Prior to initial biopsy: Tools that meet most or all of the above criteria based on available data include the Prostate Health Index (PHI), SelectMDx, 4Kscore, ExoDx Prostate Test, MyProstateScore (MPS), MPS 2.0 (MPS2), Stockholm3, and IsoPSA.
- After initial negative biopsy: Tools that meet most or all of the above criteria based on available data include percent-free PSA, 4Kscore, PHI, ConfirmMDx, ExoDx Prostate Test, MPS, MPS2, and IsoPSA.
- The Panel encourages researchers to perform randomized controlled trials that validate the safety and efficacy of biomarker tests; that compare biomarker tests to mpMRI; and that compare biomarker tests head-to-head.

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ABBREVIATIONS

AIP	**atypical intraductal proliferation**
AUC	**area under the curve**
DRE	**digital rectal examination**
H&P	**history and physical**
IDC	**intraductal carcinoma**
IQR	**interquartile range**
mpMRI	**multiparametric MRI**
MPS	**MyProstateScore**
PCPT	**Prostate Cancer Prevention Trial**
PHI	**Prostate Health Index**
PIN	**prostatic intraepithelial neoplasia**
PSA	**prostate-specific antigen**
ROC	**receiver operating characteristic**
TRUS	**transrectal ultrasound**

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

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National Comprehensive Cancer Network®

NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

Discussion

This discussion corresponds to the NCCN Guidelines for Prostate Cancer Early Detection. Last updated: February 18, 2026

Table of Contents

Summary of NCCN Recommendations for Early Detection MS-2
Overview MS-2
Guidelines Update Methodology MS-3
Literature Search Criteria and Guidelines Update Methodology MS-3
Sensitive/Inclusive Language Usage MS-3
PSA-Based Early Detection Testing MS-3
- Factors Affecting PSA Levels MS-4
- Controversies of PSA Testing MS-5

Population-Based Screening Studies MS-5
- ERSPC Trial MS-6
- PLCO Trial MS-7
- CAP Trial MS-8
- ProScreen Trial MS-8

Practical Considerations of Testing MS-8
- Age at Which to Initiate Testing MS-8
- Frequency of Testing MS-9
- Age at Which to Discontinue Testing MS-11
- Transgender and Gender-Diverse Populations MS-12

Screening in Populations with Elevated Risk MS-13
- Family History MS-13
- Genetic Syndromes MS-13
  - Hereditary Breast and Ovarian Cancer Syndrome MS-14
  - Lynch Syndrome MS-15
  - Li-Fraumeni Syndrome MS-15
- African Ancestry MS-15
- Recommendations for Early Detection in Populations with Elevated Risk MS-16

Chemoprevention MS-16
Further Evaluation and Indications for Biopsy MS-17
- Pre-Biopsy Workup MS-17
  - PSA Density MS-18
  - Risk Calculators MS-18

Magnetic Resonance Imaging MS-18
- Biparametric MRI MS-22
- Micro-Ultrasound MS-23

DRE MS-23
- DRE Alongside PSA Testing MS-23
- DRE in Patients with Elevated PSA MS-24

Biomarker Testing MS-24
- %fPSA MS-25
- PHI MS-25
- 4Kscore MS-26
- ConfirmMDx MS-26
- ExoDx Prostate (IntelliScore) MS-27
- SelectMDx MS-28
- IsoPSA MS-28
- MyProstateScore MS-29
- MyProstateScore 2.0 ..... MS-29
- Stockholm3 ..... MS-30
- Additional Biomarker Tests MS-31

Biopsy Technique MS-31
- Initial Biopsy MS-31
- Targeted Biopsy Techniques MS-31
- Repeat Biopsies MS-32
  - Targeted Biopsy for Repeat Biopsy MS-32
  - Saturation Biopsy Techniques MS-33
- Risks of Biopsy MS-33
- Management of Biopsy Results MS-35
  - Cancer MS-35
  - Intraductal Carcinoma without Invasive Cancer MS-35
  - Atypical Intraductal Proliferation without Invasive Cancer MS-35
  - Other Notable Pathologic Features and Benign Results MS-35

Summary MS-36
References MS-38

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NCCN Guidelines Version 2.2026
Prostate Cancer Early Detection

Summary of NCCN Recommendations for Early Detection

The Panel believes that most individuals with a prostate should participate in an early detection program for prostate cancer. This recommendation is particularly strong in those at higher risk (positive family history, African ancestry, and/or presence of inherited genetic variants). Patients with a shorter life expectancy (<10 to 15 years), however, are unlikely to benefit from prostate cancer early detection. The risks and benefits associated with the early detection and treatment of prostate cancer should be at least briefly explained to the patient. A decision to proceed with early detection should be made in the context of the patient's goals of care (eg, shared decision making).

Overview

Prostate cancer is the most commonly diagnosed cancer, contributing to 31% of all new cancer diagnoses in males, and is the second leading cause of cancer deaths in American men.[1] In 2026, it is estimated that 333,830 Americans will be diagnosed with prostate cancer and 36,320 will die of this disease.[1] Of concern are recent increases in the incidence of metastatic disease in those <55 years and regional/metastatic disease in individuals aged 55 to 69 years.[2]

Each year, nearly 20 million individuals in the United States will be confronted with important decisions regarding early detection for prostate cancer. Men in the United States have about 1 chance in 8 (12.8%) of eventually being diagnosed with this malignancy and about 1 chance in 49 (2.0%) of eventually dying of it.[3] Thus, prostate cancer is often not fatal. In fact, prostate cancer represents a spectrum of disease that ranges from non-aggressive, slow-growing disease that may not require treatment to aggressive, fast-growing disease that does.

The NCCN Prostate Cancer Early Detection Panel supports the use of prostate-specific antigen (PSA) testing for the early detection of prostate cancer in informed, healthy individuals with a prostate who are within certain age groups. The Panel bases this recommendation on level I evidence from randomized trials that showed a reduction in prostate cancer-specific mortality in those who underwent PSA screening. However, the Panel also uniformly acknowledges the risk of overdetection of otherwise indolent disease and the attendant risk of overtreatment, which exposes patients to the potential morbidity of treatment without benefit. Therefore, these guidelines highlight several techniques designed to improve the identification of significant cancer while avoiding the detection of indolent disease. The Panel also concludes that these NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Prostate Cancer Early Detection should be used in conjunction with the NCCN Guidelines for Prostate Cancer (available at www.NCCN.org), which explicitly recommend active surveillance or observation for appropriate candidates.

Thus, the NCCN Guidelines® for Prostate Cancer Early Detection provide a set of sequential recommendations detailing an early detection and evaluation strategy for maximizing the detection of prostate cancer that is effectively treatable and that, if left undetected, represents a risk to the patient. At the same time, these guidelines focus on minimizing unnecessary procedures and limiting the detection of indolent disease. The overall goals of these guidelines are to reduce suffering and death from prostate cancer while also limiting morbidity from unnecessary biopsies, overdiagnosis, and overtreatment.

These guidelines are intended for individuals with a prostate without a history of prostate cancer and the clinicians who care for these individuals. While shared decision making continues to be recommended, the Panel believes that the risks associated with prostate cancer early detection

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programs have decreased significantly in the past several years, especially with the use of mpMRI. Therefore, the Panel supports an early detection program in most healthy patients. Any clinician who uses these guidelines is expected to exercise independent medical judgment in the context of individual clinical circumstances, and to fully incorporate patient preferences in deciding how to apply these guidelines.

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria and Guidelines Update Methodology

Prior to the update of this version of the NCCN Guidelines for Prostate Cancer Early Detection, an electronic search of the PubMed database was performed to obtain key literature in the field of prostate cancer early detection using the following search terms: (prostate cancer) AND (screening OR early detection). The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[4]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase II; Phase III; Clinical Trial, Phase IV; Guideline; Practice Guideline; Randomized Controlled Trial; Meta-Analysis; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[5] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms *men, women, female, and male* when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

PSA-Based Early Detection Testing

PSA is a glycoprotein secreted by prostatic epithelial cells, and its protease activity lyses the clotted ejaculate to enhance sperm motility. Although primarily confined to the seminal plasma, PSA enters the circulation through unknown mechanisms. Many commercially available sources of PSA antibodies for serum tests are available worldwide. With the exception of minor differences in the calibration of these assays, they perform comparably when used appropriately. However, PSA measures obtained using different commercial assays are not directly comparable or interchangeable, since the values are calibrated against different standards. If an abnormally high PSA is observed, repeat testing should

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be performed, particularly if the value is close to the threshold value that prompts evaluation. One study showed that approximately 25% of individuals with initial PSA levels between 4 and 10 ng/mL had normal PSA values upon repeat testing.[6]

PSA is not a cancer-specific marker, and as such most individuals with elevated PSA levels do not have prostate cancer. The risk of prostate cancer increases with increasing PSA, but there is no level of PSA below which the risk of prostate cancer is zero. Total PSA (tPSA) levels >10 ng/mL confer a >67% likelihood of biopsy-detectable prostate cancer, whereas about 18% of patients with PSA in the 4 to 10 ng/mL range have a subsequent positive biopsy.[7,8] Using data from 18,882 participants in the Prostate Cancer Prevention Trial (PCPT), it was demonstrated that 15% of those with a PSA level of ≤4.0 ng/mL and a normal digital rectal examination (DRE) had prostate cancer (as diagnosed by end-of-study biopsies).[9] The PCPT investigators determined the sensitivity and specificity of PSA levels for detecting any prostate cancer using various cut-offs. At 3.1 ng/mL, PSA has a sensitivity of about 32% and a specificity of about 87%.[10]

Overall, appropriate use of PSA testing alone can provide a diagnostic lead-time of 5 to 10 years, but the lead-time varies across studies, populations, and screening protocols.[11]

Despite its limitations, population-based prostate cancer screening studies have demonstrated survival benefits using PSA—sometimes in combination with DRE or other ancillary tests— as discussed in more detail below.

Factors Affecting PSA Levels

PSA can be elevated due to infection, recent instrumentation, ejaculation, or trauma. However, empiric antibiotic use appears to have little value for improving test performance in asymptomatic patients with an elevated PSA.[12]

The 5α-reductase inhibitors (5-ARIs) finasteride and dutasteride are commonly used to treat lower urinary tract symptoms due to benign prostatic hyperplasia (BPH). Use and duration of 5-ARI therapy should be elicited carefully in the medical history, because this class of drugs typically results in an approximate 50% decrease in serum PSA levels within 6 to 12 months of initiating therapy.[13] However, this effect is tremendously variable. For example, one study showed that after 12 months of treatment, only 35% of patients demonstrated the expected 40% to 60% decrease in PSA, while another 30% had a >60% decrease.[14] Thus, the commonly used method of doubling the measured PSA value to obtain an adjusted value may result in unreliable cancer detection.

In fact, if a significant PSA decrease does not occur while taking 5-ARIs, it can indicate a heightened risk for prostate cancer that warrants regular testing. Results from several clinical trials suggested that 5-ARIs enhance the predictive capacity of PSA.[15-17] Although reflex ranges for PSA among patients on 5-ARIs have not been established, a confirmed rise from post-5-ARI treatment nadir may be a better indication for biopsy than doubling the PSA level.

It should also be noted that finasteride at 1 mg/day used to treat male-pattern baldness also lowers serum PSA.[13]

Ketoconazole, commonly used to treat fungal conditions, inhibits the androgen synthesis pathway and hence can also lower PSA levels. Since moderate PSA decreases have been observed with ketoconazole in the treatment of patients with prostate cancer,[18] recent ketoconazole use should also be noted in the medical history.

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Over 50% of men in the United States reported using dietary supplements (eg, vitamins, minerals, botanicals) in the 2017-2018 National Health and Nutrition Examination Survey.[19] It has been speculated that some dietary supplements, especially certain antioxidants, may affect serum PSA levels.[20] However, very little is known about the exact composition of these herbal supplements and their specific effects on serum PSA levels.

Controversies of PSA Testing

Widespread PSA screening, which began in the late 1980s, played a critical role in the dramatic downward migration of prostate cancer stage at diagnosis seen through the 1990s.[21,22] This trend likely contributed to a substantial reduction in prostate cancer mortality. Based on its perception of the harm-benefit tradeoffs of prostate cancer screening, the U.S. Preventive Services Task Force (USPSTF) recommended against routine PSA testing in 2012.[23] After this recommendation, prostate cancer screening decreased, as did biopsy rates, diagnoses of localized prostate cancers, and radical prostatectomy rates.[24-35] The effect of the 2012 USPSTF recommendations on the rate of metastatic prostate cancer diagnoses is, however, unclear, with some studies showing an increase and others showing none.[29,30,36-38] The Panel agrees with widespread views that, in retrospect, the 2012 USPSTF recommendation was overly influenced by a misleading interpretation of a single study and lacked the insight of longer-term follow-up of other clinical trials.[39-45]

The USPSTF released updated recommendations in 2018 based on an evidence report and systematic review.[46,47] The recommendations are: 1) against PSA-based prostate cancer screening in individuals aged ≥70 years; and 2) for individualized, informed decision-making regarding prostate cancer screening in individuals aged 55 to 69 years. For those in the latter age group, clinicians should inform them regarding the potential harms and benefits of PSA-based screening. The USPSTF statement does not provide guidance for people <55 years. Data suggest that PSA testing rates increased after this update, reversing previous trends.[48]

Although prostate cancer is a major cause of death and disability in the United States, concerns remain that early detection often results in overdetection, which is the identification of disease that would not be a problem for the patient if undetected or untreated and that would not have been identified without screening. These arguments hold that overdetection may lead to overtreatment, which is aggressive treatment in individuals with a low probability of yielding clinical benefit. However, analyses of trends in prostate cancer management show that the rates of active surveillance for early-stage disease have increased significantly, allaying initial concerns about overtreatment.[49] Moreover, modern detection methods, especially the incorporation of MRI before biopsy, have reduced detection of low-grade prostate cancer (see *Magnetic Resonance Imaging*, below).

Another possible concern is that PSA testing often produces false-positive results, which in turn contribute to patient anxiety and the increased costs and potential complications associated with unnecessary biopsies. An important modern strategy strongly endorsed by the Panel is to use MRI prior to biopsy, which can reduce unnecessary biopsies by 30% to 50%, as discussed in detail below.

Thus, the decision about whether to pursue early detection of prostate cancer is complex. When, who, and how often to test remain major topics of debate.[50-55]

Population-Based Screening Studies

Several randomized trials have assessed PSA-based prostate cancer early detection programs, as discussed in detail below. However, it is important to note that the trials with long-term follow-up were conducted in

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the pre-MRI era. MRI reduces overdiagnosis, which was a concern raised by results of these studies.

There are also at least three additional limitations of these randomized controlled trials. First, they did not address the potential benefit of screening in those with high-risk factors. For instance, <5% of participants in the Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer screening trial were of African descent and only 7% reported a family history of prostate cancer.[56] Therefore, it is not known whether individuals at higher risk may benefit more from screening than those at lower risk. Second, many individuals in these studies underwent sextant prostate biopsies rather than extended core biopsies, the standard diagnostic technique used today. Third, the ERSPC trial may have underestimated benefit due to advanced age at first PSA test (median >60 years), low intensity of screening (largely every 4 years) and suboptimal treatment available in Europe in the 1990s compared to what is available today.

ERSPC Trial

The ERSPC trial involved about 182,000 participants between the ages of 50 and 74 years in seven European countries, randomly assigned to a group that was offered PSA screening at an average of once every 4 years or to a control group that did not receive such screening; DRE or other ancillary tests were also performed in the screening group.[57,58] The predefined core group included 162,388 participants aged 55 to 69 years. Death from prostate cancer was the primary outcome. During a median follow-up of 11 years, the cumulative incidence of prostate cancer was 7.4% in the screening group versus 5.1% in the control group. There were 299 prostate cancer deaths in the screening group compared to 462 in the control group. The rate ratio for death from prostate cancer was 0.79 for the screening arm compared to the control arm (95% CI, 0.68–0.91; P = .001). The investigators concluded that the PSA-based screening program reduced mortality from prostate cancer by 21%. At the time of publication, the authors stated that 1055 individuals would need to be screened and 37 additional individuals would need to be treated over 11 years to prevent one prostate cancer death. Modeling of the ERSPC data, however, estimated that the number needed to screen was 98 and the number needed to treat was 5 to prevent one prostate cancer death.[59]

A report of 13-year follow-up of the ERSPC trial, with 7408 participants diagnosed with prostate cancer in the screening arm and 6107 diagnosed in the control arm, confirmed these results.[60] The unadjusted rate ratio for death from prostate cancer was 0.79 (95% CI, 0.69–0.91) at 13 years. After adjusting for non-participation, the rate ratio of prostate cancer death was 0.73 (95% CI, 0.61–0.88). The authors reported that, for 781 individuals invited for screening or 27 additional prostate cancers detected, one prostate cancer death could be averted. Furthermore, another analysis of these 13-year data found that fewer participants were diagnosed with metastatic disease in the screening arm (incidence rate ratio, 0.60; 95% CI, 0.52–0.70).[61] After longer follow-up (16 years), the number invited for screening and the number of prostate cancers detected to avert one prostate cancer death were reduced to 628 and 18, respectively.[62] After a median 23 years of follow-up, the corresponding numbers were 456 and 12.[42]

The apparent risk reduction was also confirmed in an analysis of the Rotterdam section of the ERSPC trial where prostate cancer-specific mortality was reduced by 32%.[63] This same group found that if one controlled for contamination and nonattendance, the risk of death due to prostate cancer could be reduced by up to 51%.[64] Twenty-one-year follow-up of the Rotterdam section found fewer advanced prostate cancer diagnoses and less disease progression, secondary treatment, and palliative treatment in the screening arm compared with the control arm.[43] Although an increase in overdiagnosis was also seen, the use of

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expectant management approaches increased over time, mitigating the amount of over-treatment.

The Finnish Prostate Cancer Screening Trial, the largest component of ERSPC, reported a small, non-statistically significant reduction in prostate cancer-specific death after 12 years of follow-up.[65]

The Göteborg randomized, population-based, prostate cancer screening trial was initiated before and independently of the ERSPC, but some of its patients were reported as part of the ERSPC.[66] Twenty thousand individuals aged 50 to 64 years were randomized to either a screening group invited for PSA testing every 2 years until the age of 69 years or to a control group. In those randomized to screening, 76% attended at least one test. PSA testing in the general population was very low at the beginning (3%) but increased over time. During a median follow-up of 14 years, 1138 participants in the screening group and 718 in the control group were diagnosed with prostate cancer, resulting in a cumulative prostate cancer incidence of 12.7% in the screening group and 8.2% in the control group (HR, 1.64; 95% CI, 1.50–1.80; $P < .0001$). The rate ratio for death from prostate cancer was 0.56 (95% CI, 0.39–0.82; $P = .002$) in the screening group compared with the control group. Overall, 293 individuals needed to be screened and 12 needed to be diagnosed to prevent one prostate cancer death over 14 years. This study shows that prostate cancer screening is acceptable to the Swedish population and that prostate cancer mortality was reduced almost by half over 14 years. In addition, it should be noted that a cause-specific survival benefit was noted despite the fact that not all cancers were immediately treated. This result suggests that early detection combined with selective treatment based on risk can lower mortality rates without uniform treatment of all cancers.

Eighteen-year follow-up of the Göteborg trial was reported, with 1396 participants diagnosed with prostate cancer in the screening arm and 962 diagnosed in the control arm.[45] The reduction in absolute prostate cancer-specific mortality was 0.72 (95% CI, 0.50–0.94). The number needed to invite to prevent one death was 139 and the number needed to diagnose was 13.

There are several possible explanations for the more favorable results of the Göteborg trial compared to the PLCO (see below) or ERSPC trials. First, the patients were younger and less likely to have incurable prostate cancer at first screening; second, there was less contamination of the control arm because PSA testing was uncommon in the Swedish population when the study began; third, a lower PSA threshold was used for recommending a biopsy; and finally, participants were screened more frequently than in ERSPC and for a longer period than in PLCO. However, because more than half of the participants were included in the main analysis of ERSPC, the Göteborg trial should not be interpreted as a true independent confirmatory study. An analysis of the Göteborg trial showed that the risks of aggressive prostate cancer and prostate cancer mortality became similar in the screening and control arms 9 years after screening cessation.[67]

PLCO Trial

The PLCO study randomized 76,685 individuals aged 55 to 74 years at 10 U.S. study centers to annual screening (annual PSA for 6 years and DRE for 4 years) or usual care.[68] After 13 years of follow-up, the incidence rate ratio for the screening arm compared to the control arm was 1.12 (95% CI, 1.07–1.17). The investigators found no statistically significant difference between the disease-specific mortality rates of the screening and control groups (relative risk [RR], 1.09; 95% CI, 0.87–1.36). Results were similar after 15-year follow-up.[69]

Despite the large sample size, this trial was flawed both by prescreening and the high contamination rate of 40% to 52% per year in the control

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group (ie, 74% of participants in the usual care arm were screened at least once). The high contamination rates have been confirmed by others.[70,71] The estimated mean number of screening PSAs (DREs) was 2.7 (1.1) in the control arm and 5.0 (3.5) in the screened arm. In addition, the biopsy rate for those with elevated serum PSA values was relatively low compared to the European trials. The PLCO trial thus really compared fixed screening versus "opportunistic" screening and, therefore, did not really test the hypothesis that screening with PSA is of value. However, it showed that yearly screening may be of limited value compared to less frequent testing.[72]

An analysis, which endeavored to account for the increased screening and diagnostic workup in the control arms of the PLCO and ERSPC, found that PSA screening lowered the risk for prostate cancer death in both trials by similar amounts (by an estimated 25%–31% in PLCO and by an estimated 27%–32% in ERSPC).[73]

In a subset analysis of PLCO, a 44% decrease in the risk of prostate cancer-specific death was observed in those with no or minimal comorbidity assigned to screening compared to control, and the numbers needed to screen and treat to prevent one death were 723 and 5, respectively.[74] This benefit was not found among individuals with one or more significant comorbidities. These results suggest that screening is more useful among individuals in good health due to the lack of competing cause for mortality. However, others suggest that such analysis is prone to major methodologic errors.[75]

CAP Trial

The results of the Cluster Randomized Trial of PSA Testing for Prostate Cancer (CAP) were reported in 2018.[76] Participants aged 50 to 69 years (n = 419,582) were randomized to a single PSA test or no screening. After a median follow-up of 10 years, 549 participants died of prostate cancer in the intervention group versus 647 in the control group (P = .50). Not surprisingly, more low-risk cancers were identified in the intervention group. No difference in all-cause mortality was seen. In a median 15-year follow-up analysis of the same trial, however, a statistically significant reduction in prostate cancer mortality of 0.09% compared to the control group was evident (0.69% vs 0.78%; P = .03).[44] Although this trial had several very important strengths, it had limitations as well. Only a single PSA test was used, a standard 10-core biopsy was undertaken, there was very little representation of Black individuals, and only 40% of participants completed the intervention (biopsy). Serial testing, higher rates of biopsy completion, longer follow-up, and use of additional technology preceding biopsy (discussed below) may have led to greater benefit with PSA testing.

ProScreen Trial

The ProScreen trial, which randomized 60,745 individuals to compare the incidences of Grade Group 1 and Grade Group ≥2 prostate cancers, utilized a 1:3 allocation to either an invitation for screening (intervention) or no invitation for screening (control) group.[4] If PSA level of 3 ng/mL was detected, participating individuals would then undergo 4K risk score testing (see *Biomarker Testing*, below). 4K risk scores ≥7.5% led to individuals undergoing prostate gland MRIs, followed by targeted biopsy if MRI-positive. After a 3.2-year median follow-up period, the number of detected lower-grade and high-grade prostate cancers were greater for the invited screening group (0.26% and 1.13%, respectively) compared to those in the non-invited group (0.14% and 0.62%, respectively).

Practical Considerations of Testing

Age at Which to Initiate Testing

Controversy exists as to the ideal age to begin screening for prostate cancer. Randomized trials looking at the impact of screening on prostate cancer mortality have focused primarily on individuals aged 55 to 69

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years. The ERSPC and Göteborg trials reported decreased disease-specific mortality in individuals aged 55 to 69 and 50 to 64 years, respectively. These results support baseline PSA testing in individuals aged 50 to 55 years with the strongest evidence supporting testing at age 55 years. Analyses of PSA testing in Swedish individuals aged 50 to 54 years support screening in this age group.[77]

As individuals <50 years were not included in these screening studies, baseline testing at earlier ages has not been evaluated in RCTs. However, observational evidence suggests that baseline testing of individuals in their 40s and early 50s may have value for future risk stratification, although some would describe the value as marginal.[78] One study assessed blood collected from 21,277 participants in Sweden aged 33 to 50 years who were followed until 2006.[79] Among the 1312 patients with prostate cancer and 3728 control participants without prostate cancer, these investigators reported that a single PSA test before age 50 years predicted subsequent prostate cancer up to 30 years later with a robust area under the curve (AUC) of 0.72 (0.75 for advanced prostate cancer). However, the possible risks of unnecessary biopsies and prostate cancer overdetection should be acknowledged with earlier initiation of screening.[80]

Another report clarified associations of age with the long-term risks of metastases.[81] In this study, the risk of prostate cancer death was strongly correlated with baseline PSA in individuals aged 45 to 49 years and 51 to 55 years; 44% of the deaths in the analytic cohort occurred in participants in the highest tenth of the distribution of PSA, suggesting that there may be a strong rationale for baseline testing in those <55 years.

In a nested case-control study of individuals 40 to 59 years of age in the Physicians' Health Study, baseline PSA strongly predicted lethal prostate cancer later in life.[82] For example, those aged 55 to 59 years with PSA levels above the 90th percentile had an odds ratio (OR) of 6.9 (95% CI, 2.5–19.1) for lethal prostate cancer compared with those whose PSA levels were at or below the median. A secondary analysis of a cohort in the PLCO trial also found a strong correlation between baseline PSA levels in participants aged 55 to 60 years and the actuarial 13-year incidence of clinically significant prostate cancer.[83] Those with baseline PSA levels of ≤0.49 ng/mL, 0.50–0.99 ng/mL, 1.00–1.99 ng/mL, 2.00–2.99 ng/mL, 3.00–3.99 ng/mL, and ≥4.00 ng/mL had incidence rates of 0.4% (95% CI, 0%–0.8%), 1.5% (95% CI, 1.1%–1.9%), 5.4% (95% CI, 4.4%–6.4%), 10.6% (95% CI, 8.3%–12.9%), 15.3% (95% CI, 11.4%–19.2%), and 29.5% (95% CI, 24.2%–34.8%), respectively (all pairwise log-rank $P \leq .004$).

Taken together, these results suggest that one could perform early baseline testing and then determine the frequency of testing based on risk. Although many physicians advocate earlier testing only in individuals thought to be at higher risk due to family history or African ancestry, a baseline serum PSA is a stronger predictor of the future risk of the disease compared to either of these risk factors.

Most Panel members favor informed testing beginning at age 45 years for patients with an average risk of developing prostate cancer. Patients considered at high risk for the development of prostate cancer (Black/African American individuals, those with germline variants linked to the development of prostate cancer, and those with a concerning family or personal history) should consider shared decision-making regarding early detection at age 40. Repeat testing intervals are discussed below.

Frequency of Testing

The ideal screening interval to maximize mortality reduction yet minimize overdiagnosis remains uncertain.

A comparison of two centers involved in the ERSPC trial studied the impact of different screening intervals on the diagnosis of interval cancers in participants aged 55 to 64 years.[84] The Göteborg arm randomized 4202

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participants to screening every 2 years, while the Rotterdam arm randomized 13,301 participants to screening every 4 years with similar follow-up of 11 to 12 years. Compared to screening every 4 years, there was a significant 43% reduction in the diagnosis of advanced prostate cancer (clinical stage >T3a, N1, or M1; PSA >20 ng/mL; Grade Group 5 at biopsy) for screening every 2 years. However, there was also a 46% increase in the diagnosis of low-risk prostate cancer (clinical stage T1c, Gleason <6, and PSA <10 ng/mL at biopsy) for screening every 2 years.

Some studies have addressed the effect of PSA levels on the interval of testing, and it appears that individuals with a very low PSA could safely extend the testing interval. In the Rotterdam section of the ERSPC trial, participants with a PSA <1 ng/mL had a very low risk for cancer at 4 and 8 years (0.23% and 0.49%).[85] Other studies have shown that PSA values at younger ages strongly predict the development of or death from prostate cancer.[86,87] For example, in a Swedish case-control study of 1167 individuals, those aged 60 years with PSA concentrations of ≤1 ng/mL had only a 0.5% risk of metastasis by age 85 and a 0.2% risk of death from prostate cancer.[87] Other analyses, some with racially and ethnically diverse cohorts, show a similar correlation with a very low risk of prostate cancer diagnoses, metastases, and death for those with PSA levels in the lower percentiles for their age.[81-83,86,88,89] Overall, the available data show that individuals who have a PSA above the median for their age group are at a higher risk for prostate cancer and aggressive prostate cancer. The higher above the median, the greater the risk.

In another study, the unscreened Malmö cohort was compared to a cohort with regular screening.[90] The authors concluded that the harms of testing outweigh the benefits for those with PSA <1 ng/mL at age 60 years (approximately 50% of the population). The authors estimated that PSA testing every 2 years in individuals with a PSA <1 ng/mL would result in 171 additional diagnoses per 10,000 individuals in 15 years without a concomitant reduction in prostate cancer mortality.

Microsimulation models suggest that PSA-based screening intervals (with less frequent screening in those with low PSAs) could substantially reduce the testing burden while preserving the lives saved.[91] Another study using microsimulation models of prostate cancer incidence and mortality predicted that a strategy that utilizes biennial intervals for those with average PSA levels and longer screening intervals (every 5 years) for those with low PSA levels (below median for age by decade) allows a 2.27% risk of prostate cancer death compared to 2.86% from no screening.[92] In addition, compared to annual screening and using a biopsy threshold of 4.0 ng/mL, the biennial strategy also projected a relatively lower overdiagnosis rate of 2.4% (vs 3.3% for annual screening), a 59% reduction in total tests, and a 50% reduction in false-positive results. The biennial model was robust to sensitivity analyses, which varied the range of cancer incidence and survival attributed to screening.

There is also some evidence that shorter testing intervals may be justified in patients who are at higher risk for the development of prostate cancer. For example, a modeling study looked at the effects of different PSA intervals in Black individuals.[93] The study showed that a shorter interval could reduce mortality without increasing the risk of overdiagnosis in Black people.

After considering these data, the Panel concluded that tailoring screening intervals based on PSA and risk levels might maximize survival advantage while decreasing the number of screenings and limiting overdiagnosis. The Panel recommends repeat testing at 2- to 4-year intervals for individuals with average risk and a PSA <1 ng/mL. For those with high risk and PSA ≤3 ng/mL and those with average risk and PSA 1–3 ng/mL, 1- to 2-year intervals are recommended. Further evaluation can be considered for younger patients and for those who have PSA levels that are above the

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median for their age group, even if they do not meet the threshold of >3 ng/mL. For patients >75 years of age, repeat testing is recommended at 1- to 3-year intervals if PSA is <4 ng/mL, but only in very healthy people with little or no comorbidity. Discontinuation of screening can be considered if clinically appropriate. Clinical judgment and shared decision-making should be used to determine the testing interval within these ranges for each individual.

Age at Which to Discontinue Testing

Panelists uniformly agree that PSA testing should not be offered to individuals with a life expectancy of <10 years. Several resources exist to help clinicians estimate life expectancy, such as the Social Security Administration's Actuarial Life Table, the WHO's Life Tables, and the Memorial Sloan Kettering's Male Life Expectancy tool. However, the ideal age at which to discontinue screening for asymptomatic individuals with normal PSA levels is unknown. A balance is needed between limiting unnecessary biopsies/overdiagnosis and maximizing the detection of aggressive disease that could threaten the lives of older individuals.

Since the previously cited RCTs (ERSPC, PLCO, and Göteborg) observed benefits to testing only in participants aged ≤70 years, several Panel members favored stopping testing at age 70 years.

However, other data would suggest a benefit to screening beyond 70 years. A study of 4561 patients who underwent radical prostatectomy found that those >70 years were more likely to have higher grade and stage of disease and worse survival compared to those ≤70 years.[94] Others have published similar findings.[95] In addition, an analysis of the SEER database showed that patients who were diagnosed with prostate cancer at >70 years of age were more likely to die of their disease than those diagnosed at <70 years (21% vs 17%).[96] A separate analysis of the SEER database showed that patients whose prostate cancer was diagnosed at ≥75 years were more likely to have advanced disease and had a higher risk of death from prostate cancer despite greater death rates from competing causes than those diagnosed at <75 years.[97] Furthermore, a cross-sectional population study in Norway showed that the absolute incidence of metastatic disease continues to rise with age.[98] Another cohort study in Sweden also showed effects of age on prostate cancer death, and further showed that patients who are older may not receive sufficient diagnostic workup and curative treatment, thus confounding the interpretation of the results.[99] Another possible issue with the interpretation of these studies is that fewer of the patients >70 years may have undergone PSA-based early detection than in the <70-year-old population, explaining the higher prevalence of advanced disease at diagnosis.

To assess the appropriate ages for discontinuing screening, the previously cited microsimulation model[92] predicted that decreasing the stopping age from 74 to 69 years would lead to a 27% relative reduction in the probability of life saved, but to an almost 50% reduction in the probability of overdiagnosis. This latter finding reflects the fact that a large proportion of individuals >70 years have cancer that would be unlikely to diminish their life expectancy, and that screening in this population would substantially increase rates of overdetection, while also recognizing the increased prevalence of higher-risk disease in this age group that could benefit from earlier detection. It is also unclear how these expected rates of overdetection would be modified by use of pre-biopsy MRI.

The microsimulation model also assessed a strategy of screening individuals up to age 74 years while simultaneously increasing the PSA threshold for biopsy based on age-dependent PSA levels (ie, increasing the threshold level for biopsy with increasing age). Compared to using a uniform cutoff of 4.0 ng/mL, this strategy reduced the rate of overdiagnosis by one third while only slightly altering lives saved.

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PSA at certain ages may predict future risk. One study examined the relationship between baseline PSA at age 60 years and the future risk of prostate cancer death or metastases and found that those with a PSA level below the median (<1 ng/mL) were unlikely to develop clinically significant prostate cancer (0.5% risk of metastases and 0.2% risk of prostate cancer death). [87] Similarly, in a study of 849 individuals in the Baltimore Longitudinal Study of Aging (BLSA), no individuals aged 75 to 80 years with a PSA <3.0 ng/mL died of prostate cancer.[100] Moreover, the time to death or diagnosis of aggressive prostate cancer was longer in those with a PSA <3.0 ng/mL versus those with a PSA >3.0 ng/mL, suggesting that individuals ≥75 years with a PSA <3.0 ng/mL are unlikely to die or experience aggressive prostate cancer throughout their remaining life and most may safely discontinue screening.

In summary, many possible strategies to reduce overdiagnosis in the older population exist. Individuals ≥60 years with a PSA <1.0 ng/mL and those >75 years with a PSA <3.0 ng/mL have a very low risk of prostate cancer metastases. Continuing screening beyond age 75 years should be performed only with caution in very healthy patients with little to no comorbidity, especially if they have never undergone PSA testing or have increasing PSA levels (category 2B for continuing screening beyond age 75 years), to detect the small number of aggressive cancers that pose a significant risk if left undetected until signs or symptoms develop. Widespread testing in this population would substantially increase rates of overdetection and is not recommended. Individuals >75 years who choose to continue PSA-based prostate cancer early detection (category 2B) and who have a PSA <4 ng/mL, a normal DRE (if done), and no other indications for biopsy can undergo repeat testing at 1- to 3-year intervals (or consider discontinuing screening if clinically appropriate), but again only in very select patients. Those with a PSA ≥4 ng/mL and/or a DRE that is very suspicious for cancer should undergo further evaluation as indicated in the guidelines.

Transgender and Gender-Diverse Populations

Transgender women and other gender-diverse individuals with a prostate are a growing segment of the population who warrant special consideration. The prostate is not removed in any form of gender-affirmation surgery, and all individuals with prostates remain at risk of prostate cancer. However, rates of prostate cancer in transgender women appear to be lower than that in cisgender males.[101] The largest published case series on prostate cancer in transgender women to date (N = 155) showed that patients on gender-affirming hormones had more aggressive disease at diagnosis, suggesting potential delays in diagnosis of clinically significant disease.[102] Thus, clinicians should engage transgender women and gender-diverse individuals with a prostate in shared decision-making regarding age-appropriate PSA screening.

For those on gender-affirming hormones, such as estrogen, PSA values should be interpreted cautiously as existing reference ranges do not account for the castrating effects of gender-affirming hormones and are likely not appropriate for use in this population. A retrospective study found that the median PSA level was 0.02 ng/mL (interquartile range [IQR], 0–0.2; 95th percentile, 0.6 ng/mL) in 210 transgender women with a median age of 60 years who were on estrogen for at least 6 consecutive months and had no known prostate cancer.[101] The study reported that similarly aged cisgender men without prostate cancer have a median PSA of 1 ng/mL in other studies. A systematic review published in 2025 identified 4 relevant studies and supported the conclusion of the study presented above – that individuals with prostates who are taking gender-affirming hormone therapy have mean/median PSA levels ranging from 0.02 to 0.53 ng/mL.[103]

Future dedicated studies are needed to define optimal screening practices and, specifically, to determine proper thresholds to define elevated PSA levels in transgender and gender-diverse individuals with a prostate

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undergoing gender-affirming care. In the meantime, clinicians should take the far lower PSA values into account when interpreting PSA values and pay special attention to rising PSA levels in this population. Further evaluation may be warranted even if values are within existing reference ranges.

Screening in Populations with Elevated Risk

Individuals with a concerning family or personal history, Black/African American individuals, and those with certain germline variants have a higher risk of developing prostate cancer, as discussed in further detail below.[104-110] In addition, hormone disrupters such as 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD), the carcinogenic chemical compound found in the herbicide Agent Orange, may also be linked to the development of prostate cancer.[111,112] Patients with exposure to Agent Orange (eg, many Vietnam War veterans) should be informed that exposure to this chemical may be associated with an increased risk of Grade Group ≥2 prostate cancer.

Family History

Having a first-degree relative with prostate cancer diagnosed before the age of 60 increases the likelihood of a prostate cancer diagnosis by 2.1- to 2.5-fold.[106,107] Furthermore, in individuals who have a sibling with aggressive prostate cancer, the OR for aggressive prostate cancer is 1.21 (95% CI, 1.04–1.39).[113] A population-based study in Sweden found that the risk for the development of prostate cancer increased with the number of affected relatives.[114] Data, however, suggest that prostate cancer in individuals with a family history of prostate cancer is not more likely to be aggressive, and cancer-specific outcomes are similar between those with and without a family history.[109,115,116]

Prostate cancer risk is also elevated in patients with a family history of breast cancer.[117,118] The Panel lists cancer types that should be considered during the family and personal cancer history risk assessment in the guidelines above. Patients with a suspicious family history should undergo genetic testing.

It is important to note that those with a family history of prostate cancer are more likely to undergo screening and biopsy than those without a family history. Therefore, the role of family history as a risk factor for prostate cancer may be overestimated.[119] Welch and Brawley refer to this phenomenon as “self-fulfilling risk factors” in cancers that are “scrutiny-dependent.”[120]

Genetic Syndromes

For the most up-to-date list of prostate cancer susceptibility genes and cancer risk management recommendations based on genetic testing results, see the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate (GENE-A and BRCA-A; available at www.NCCN.org). Information regarding the status of high-risk germline variants should be used as part of the discussion about prostate cancer early detection; patients may not be aware of the increased risk for prostate cancer associated with such variants. Data on the most common variants are discussed below.

Evidence indicates that patients with prostate cancer may have germline variants in 1 of 16 DNA repair genes: *BRCA2* (5%), *ATM* (2%), *CHEK2* (2%), *BRCA1* (1%), *RAD51D* (0.4%), *PALB2* (0.4%), *ATR* (0.3%), and *NBN*, *PMS2*, *GEN1*, *MSH2*, *MSH6*, *RAD51C*, *MRE11A*, *BRIP1*, or *FAM175A*.[121] Individuals with these inherited variants may have an increased risk for prostate cancer. Several inherited syndromes are discussed below. In addition, individuals who carry the rare G84E variant of the *HOXB13* gene also have a significantly higher risk for prostate cancer and are more likely to have early-onset familial disease.[122-124] The *HOXB13* variant is more frequent among families of Scandinavian

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heritage. *HOXB13* X285K is another rare variant, found in individuals of West African ancestry, which may also be associated with an increased risk of prostate cancer.[125]

The NCCN Prostate Cancer Early Detection Panel recommends inquiring about known personal or familial germline variants associated with an elevated risk of cancer. If there is a known or suspected cancer susceptibility gene, referral to a cancer genetics professional is recommended.

In addition, patients who meet hereditary risk assessment criteria established in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate and the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric (available at www.NCCN.org) should be referred for genetic counseling/testing as appropriate.[126]

Hereditary Breast and Ovarian Cancer Syndrome

Germline *BRCA1* and *BRCA2* variants associated with hereditary breast and/or ovarian cancer syndrome occur in approximately 0.2% to 0.3% of the general population, with higher rates seen in certain racial/ethnic groups.[127,128] These variants have been associated with an increased risk for prostate cancer in numerous reports.[129-138] In particular, *BRCA2* variants have been associated with a 2- to 6-fold increase in the risk for prostate cancer, whereas the association of *BRCA1* variants and increased risks for prostate cancer are less consistent.[130,132,133,138-141] Furthermore, prostate cancer in those with germline *BRCA* variants appears to occur earlier, has a more aggressive phenotype, and is associated with significantly reduced survival times than in patients who are non-carriers.[142-147]

Interim results of the IMPACT study, which prospectively enrolled participants aged 40 to 69 years with germline *BRCA1/2* variants (919 *BRCA1* carriers; 902 *BRCA2* carriers) and a control group with wild-type *BRCA1/2* who are related to carriers (709 *BRCA1* non-carriers; 497 *BRCA2* non-carriers), were reported in 2019.[148] All participants underwent annual PSA testing four times. Whereas no differences were seen between *BRCA1* carriers and non-carriers, the *BRCA2* carriers had a higher cancer incidence rate (19.4 per 1000 person years vs 12.0; P = .03); were diagnosed at a younger age (61 vs 64 years; P = .04); and were more likely to have clinically significant disease (77% vs 40%; P = .01) than *BRCA2* non-carriers. One limitation of the study is that the biopsy completion rate was higher in *BRCA2* carriers compared with non-carriers (73% vs 60%). Further follow-up is required to assess the clinical utility of this early detection strategy.

The use of mpMRI along with PSA as an upfront screening approach in germline *BRCA* variant carriers and carriers of other germline variants associated with an increased risk of prostate cancer is being studied.[149-151] Initial results suggest that the addition of mpMRI in this population may increase the sensitivity and specificity of early detection over that of PSA alone. Furthermore, the overall detection of clinically significant prostate cancer in the referenced studies ranged from 2.3% to 6.9% of the full populations and 7.6% to 33.3% of those undergoing biopsy. The NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate (available at www.NCCN.org) include consideration of a baseline mpMRI by age 50 years or at an age dependent on the prostate cancer family history, preferably on a clinical trial, in patients with germline pathogenic or likely pathogenic *BRCA* variants. For individuals considering prostate cancer screening with mpMRI, that Panel recommends that screening be performed in experienced high-volume centers. That Panel notes that such screening should only take place after an in-depth discussion about the potential

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limitations to screening, including cost, the high incidence of benign or indeterminate abnormalities, and uncertainties about the potential benefits of prostate cancer screening with mpMRI.

Lynch Syndrome

Lynch syndrome is a hereditary cancer syndrome caused by pathogenic germline variants in DNA mismatch repair genes (*MLH1*, *MSH2*, *MSH6*, or *PMS2*) or in *EPCAM*.[152] Lynch syndrome is predominantly associated with colorectal and endometrial cancers, but individuals with Lynch syndrome also have a 2- to 5.8-fold increase in risk for prostate cancer.[153-158] Age of onset and aggressiveness of prostate cancer in these individuals do not generally appear to be different than in sporadic cases.[154,157] Interestingly, a study of 524 families with pathogenic germline *PALB2* alterations found no evidence for an increased risk of prostate cancer.[159]

The ongoing international IMPACT study prospectively enrolled participants aged 40 to 69 years with germline variants in *MLH1*, *MSH2*, or *MSH6* (n = 644) and age-matched, related controls without the germline variants (n = 184).[152] An additional 134 non-carriers from the *BRCA1/2* cohort of the IMPACT study were randomly selected to be included in the control group. All participants are required to have annual PSA testing for a minimum of 5 years. After the first round of PSA screening, the incidence of prostate cancer was 4.3% among *MSH2* carriers (13 of 305; 95% CI, 2.3–7.2), 3.0% among *MSH6* carriers (4 of 135; 95% CI, 0.8–7.4), and 0.5% among *MSH2* non-carrier controls (1 of 210; 95% CI, 0–2.6). None of the *MLH1* carriers or the other non-carrier control participants were diagnosed with prostate cancer. The estimated completion of IMPACT is in the year 2030.

Li-Fraumeni Syndrome

Li-Fraumeni syndrome (LFS) is caused by a pathogenic germline variant in *TP53*. Individuals with LFS have an increased risk of many primary cancers, most commonly breast cancer, osteosarcomas, adrenocortical carcinomas, central nervous system (CNS) tumors, and soft tissue sarcomas.[160] It is also associated with an approximately 9-fold increased risk of prostate cancer.[161] Data also suggest that individuals with germline *TP53* variants are more likely to develop aggressive prostate cancer.[161]

African Ancestry

Black individuals and those with African ancestry have a 60% to 67% higher incidence of prostate cancer compared with white individuals.[162,163] Data also suggest that Black/African American individuals have an earlier onset of prostate cancer.[163-165] The higher incidence and earlier age at diagnosis are attributable to a greater incidence of pre-clinical prostate cancer and a higher likelihood that a pre-clinical tumor will spread. Although the Panel acknowledges that race is a human-made construct and thus not linked to differences in biology, they believe that there may be factors linked to African ancestry itself that may play a role.[166,167]

Data also show that Black individuals are more likely to be diagnosed with more advanced disease.[163,168-171] Furthermore, the mortality rate from prostate cancer in the Black American population is two to four times higher than all other racial and ethnic groups.[162,163,172] [163-165] The main drivers for the racial inequities are likely lower rates of prostate cancer screening, diagnostic procedures, and overall access to health care; treatment disparities; inadequate health insurance coverage; and other factors related to social determinants of health (eg, environmental exposures; patient and physician behaviors; delays in diagnosis).[162,171,173-178] A logistic regression analysis of SEER data from 2004 to 2011 determined that Black individuals also had a decreased likelihood of receiving treatment even when diagnosed with high-risk disease. [179] In fact, data showed an inverse relationship between disease severity and probability of receiving treatment in Black individuals. Further evidence supports the likelihood that such inequities are not related to race itself or

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differences in the frequency of tumor-acquired genomic variants.[180] In fact, data show that Black patients who receive equitable prostate cancer treatment have similar outcomes to white patients.[181]

Despite these inequities, prostate cancer screening has been predominantly studied in white individuals. In the PLCO trial, non-Hispanic Black participants only made up approximately 4.4% of the trial population, and 6.9% had a positive family history, but no subset analyses were performed.[56] In the ERSPC trial, no information on ancestry or family history was reported.[57]

Some data have informed how different prostate cancer early detection protocols could be used to decrease these racial inequities. For example, a study of 41,250 individuals in the Veterans Affairs (VA) Health Care System database found that the optimal PSA threshold for predicting the diagnosis of prostate cancer within 4 years was lower in African American individuals than in white American individuals (1.9 vs 2.5 ng/mL).[182] The prospective Southern Community Cohort Study found that Black individuals with PSA above the 90th percentile at ages 40 to 64 years had a greatly elevated risk of aggressive prostate cancer compared with the risk of those whose PSA levels were below the median.[89] Another analysis that estimated cases of overdiagnoses and overtreatment and calculated numbers needed to diagnose and numbers needed to treat to prevent one prostate cancer death in individuals of all races and Black individuals, found a more favorable harm-benefit ratio of PSA screening in Black individuals versus all races combined.[183] In addition, a modeling study looked at the effects of different PSA testing intervals in Black individuals and found that a shorter interval could reduce mortality without increasing the risk of overdiagnosis in this population.[93]

Recommendations for Early Detection in Populations with Elevated Risk

In conclusion, Black/African American individuals, individuals with a suspicious family or personal cancer history, and those with a known genetic predisposition represent groups at high risk for the development of prostate cancer. The Panel acknowledges that current data are insufficient to definitively inform the best strategy for prostate cancer early detection in these populations, but they recommend the following based on the data above and Panel consensus:

- It is reasonable for individuals with a suspicious family cancer history or known germline mutation that increases the risk for prostate cancer (as defined in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate; available at www.NCCN.org) to begin shared decision-making about PSA screening at age 40 years.
- It is reasonable for Black/African American individuals to consider beginning shared decision-making about PSA screening at age 40 years and to consider screening at annual rather than less frequent screening intervals. However, no current evidence shows that testing individuals of African descent at an earlier age will result in decreased morbidity and mortality compared to testing at age 45 years, and earlier screening may increase overdiagnosis.

Chemoprevention

The PCPT of 18,882 participants demonstrated that finasteride reduced the incidence of prostate cancer by 25% compared to placebo.[184] The decreased risk persisted at 21% through 16 years of follow-up.[185] This reduction was almost exclusively for low-grade (Grade Group 1) tumors; an increased proportion of aggressive (Grade Group ≥2) tumors was seen. However, after 18 years of follow-up, there was no significant difference in overall survival or survival after the diagnosis of prostate cancer in those on finasteride compared to the control group.[186] In addition, after a median

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follow-up of 18.4 years, fewer deaths due to prostate cancer were seen in the finasteride group.[187] Although this difference was not statistically significant, the results suggest that earlier fears that finasteride use would lead to an increase in prostate cancer mortality were unfounded.

In the Reduction by Dutasteride of Prostate Cancer Events (REDUCE) trial, PSA detected more high-grade Grade Group ≥2 tumors in the dutasteride arm, while the overall prostate cancer diagnosis fell by 23% compared to the control arm.[15] Similar to the PCPT trial, the difference in the number of high-grade cancers detected did not result in a mortality difference.[188]

A report on the CombAT trial also showed a 40% lower incidence of prostate cancer with dutasteride plus tamsulosin (another BPH drug) compared to tamsulosin alone, along with a slightly improved yield of PSA-driven biopsy.[17] Unlike the PCPT and REDUCE studies, diagnosis of high-grade (Grade Group ≥2) tumors was not increased.

A population-based prospective study using data from 333,820 participants in the Stockholm PSA and Biopsy Register and Prescribed Drug Register found that 5-ARI exposure decreased the risk for prostate cancer overall and for prostate cancer of Grade Group ≤3, with longer courses resulting in a larger decreased risk.[189] Grade Group 4–5 prostate cancer risk was unaffected by 5-ARI treatment.

Overall, these studies suggest that PSA testing may have enhanced specificity for patients receiving finasteride or dutasteride. However, in another population-based cohort study, researchers analyzed >80,000 records of patients with prostate cancer and found that use of 5-ARI before diagnosis was associated with delayed diagnosis, higher stage at diagnosis, higher prostate cancer–specific mortality, and higher all-cause mortality.[190] Whether or not individuals should consider taking these agents for chemoprevention is beyond the scope of this guideline.

Further Evaluation and Indications for Biopsy

The previously cited RCTs used PSA thresholds to prompt a biopsy. PSA cut points for biopsy varied somewhat between centers and trials over time. Although a serum PSA of 2.5 ng/mL has been used by many, a level of 3 ng/mL is supported by the trials and would more robustly limit the risk of overdetection. A higher threshold of 4 ng/mL is recommended for patients who choose to continue PSA screening past the age of 75 years. However, some Panel members did not recommend limiting the option of biopsy to pre-specified PSA thresholds, noting that there are many other factors (eg, age, ancestry, family history, PSA kinetics) that should also inform the decision to perform biopsy.

The Panel does not believe that DRE alone should be an absolute indication for biopsy in individuals with low PSA. The PPV of DRE in those with low PSA is poor (see *DRE*, below).[191,192] However, a DRE that is very suspicious for cancer, independent of PSA, could be an indication of clinically significant cancer in individuals with normal PSA values, and therefore biopsy can be considered. Clinical judgment should be used.

Pre-Biopsy Workup

The Panel recommends that any individual with a PSA >3 ng/mL undergo workup for benign disease, a repeat PSA, and a DRE (if not performed during initial risk assessment). An abnormal DRE in this setting of elevated PSA has a high predictive value,[193] and the Panel strongly recommends biopsy in these individuals.

The Panel strongly recommends that mpMRI (category 1), including PSA density, should precede biopsy. Biomarkers (category 2B) and/or risk calculators that improve the specificity of screening may also be considered before biopsy. The roles of imaging, PSA density, risk calculators, and biomarker testing to inform biopsy decisions are discussed in detail below. The predictive value of biomarkers has not been

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correlated consistently with that of mpMRI. Therefore, it is not known with certainty how such tests could be applied in optimal combination.

Persistent acceleration in PSA over and above an established baseline is concerning for high-risk prostate cancer.[194-196] Patients with a persistent and significant increase in PSA should be encouraged to undergo biopsy, even if they have a normal MRI.

If there is low suspicion for clinically significant cancer, individuals should be followed up in 6 to 12 months with PSA and DRE. Patients for whom there is a high suspicion of clinically significant prostate cancer should be encouraged to undergo biopsy.

PSA Density

PSA density requires the measurement of prostate volume by imaging and is expressed as the PSA value (in ng/mL) divided by prostate volume (in cc). PSA density is a means of discriminating prostate cancer from BPH: the lower the PSA density, the greater the probability of BPH.[197,198] Thus, PSA density potentially identifies individuals who do not have prostate cancer but have high PSA secondary to large-volume prostates. A PSA density cutoff of 0.15 ng/mL/g was recommended in earlier studies, which spared as many as 50% of individuals from unnecessary biopsies. This cutoff value was validated in contemporary studies and can result in a 48% reduction in unnecessary biopsies while maintaining a sensitivity of 95%.[199]

PSA density has also been shown to correlate with prostate cancer presence and aggressiveness, and may predict adverse pathology and biochemical progression after treatment.[200,201] Therefore, the Panel believes that PSA density should be considered when deciding whether to proceed to a biopsy in an individual with a negative mpMRI result or for any patients who are biopsy-naïve with an elevated PSA.

Risk Calculators

Prostate cancer risk calculators have been developed to estimate an individual's risk for prostate cancer from multiple factors. Common calculators are the Sunnybrook-, ERSPC-, and PCPT-based risk calculators.[202-209] These online tools combine clinical variables—including but not limited to age, family history, ancestry, DRE, and PSA—to estimate both the risk for biopsy-detectable prostate cancer and the risk for biopsy-detectable Grade Group ≥2 prostate cancer. Such information potentially allows for more informed decision-making.[210] However, such calculators have not been assessed in RCTs, and cut points of risk associated with reductions in prostate cancer mortality remain unknown. Such calculators have as much value in determining who might not need biopsy as in identifying those at higher risk. At this time, the Panel does not recommend the use of risk calculators *alone* to determine whether biopsy is indicated; however, they can be considered when deciding whether a biopsy can be avoided in an individual with a negative mpMRI result.

Magnetic Resonance Imaging

Considerable interest exists in using mpMRI both pre-biopsy to select patients for biopsy and during biopsy to guide needle placement.[211-218] The goals of using mpMRI in these settings include reducing the number of individuals undergoing biopsy, reducing the detection of indolent disease (and thus the risks of overdetection and overtreatment), and improving the detection of clinically significant disease through targeted biopsies.

MRI has generally been shown to have superior sensitivity for clinically significant prostate cancer when compared to TRUS-guided systematic biopsy. In the multicenter, paired-cohort PROMIS study, 576 participants with no prior biopsy and elevated PSA <15 ng/mL underwent mpMRI followed by transrectal ultrasound (TRUS) biopsy and perineal template mapping biopsy.[219] Clinically significant cancer (Grade Group ≥3 or a

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maximum cancer core length ≥6 mm by template mapping biopsy) was found in 40% of patients. In detecting clinically significant prostate cancer (some may question the cut point of ≥6 mm used), MRI was more sensitive (93%; 95% CI, 88%–96%) than TRUS biopsy (48%; 95% CI, 42%–55%; *P* < .0001), but less specific (41%; 95% CI, 36%–46% for MRI vs 96%; 94%–98% for TRUS biopsy; *P* < .0001). If a normal MRI had been used to screen individuals for biopsy, 27% would have avoided biopsy. It is important to note that patients in this study did not undergo MRI-targeted biopsy, so the study does not provide direct evidence about the performance of MRI-guided biopsy.

An approach utilizing MRI prior to biopsy followed by MRI-guided biopsy was directly compared to conventional TRUS biopsy in the PRECISION trial.[220] PRECISION was a randomized, noninferiority trial conducted in 25 centers across 11 countries. A total of 500 participants were randomized to either MRI (with or without targeted biopsy) or a 10- to 12-core TRUS biopsy. In the MRI arm, 28% avoided biopsy based on a normal MRI (Prostate Imaging Reporting and Data System [PI-RADS] <3). Clinically significant prostate cancer was detected in 38% of patients in the MRI group and in 26% in the standard biopsy group. Thirteen percent fewer participants in the MRI group received a diagnosis of low-risk disease. Because PRECISION was conducted primarily in Europe at centers of excellence in prostate MRI, the generalizability of some of its findings remain unproven. The finding that MRI before biopsy reduces negative biopsies and detection of indolent disease has been substantiated in numerous studies in both the initial and repeat biopsy settings.[221-225]

MRI-FIRST was a prospective, multicenter, paired diagnostic study conducted at 16 centers in France.[226] Each of 251 participants underwent pre-biopsy MRI, systematic biopsy, and targeted biopsy (if MRI was abnormal). Fifty-three (21%) had a normal MRI (Likert 1–2). Of these, five (11%) had clinically significant prostate cancer on systematic biopsy. Overall, clinically significant cancer (Grade Group ≥2) was found in 29.9% of patients (95% CI, 24.3–36.0) on systematic biopsy, 32.3% (95% CI, 26.5–38.4) on targeted biopsy, and 37% using combined targeted and systematic biopsy. There was no significant difference between targeted and systematic biopsy (*P* = .38). Clinically significant prostate cancer would have been missed in 5.2% of patients had systematic biopsy been skipped and in 7.6% had targeted biopsy been skipped. Thus, MRI-FIRST demonstrates that adding MRI-targeted biopsy for those without prior biopsy improves the yield of clinically significant prostate cancer, but that maximizing detection of clinically significant prostate cancer requires the combination of targeted and systematic biopsy. The study protocol allowed for different methods of targeted biopsy (cognitive and fusion), and MRIs were read locally. Still, generalizability questions remain because all study centers had experience with prostate MRI and targeted biopsy, two centers enrolled 36% of all patients, and interobserver agreement in MRI interpretation was not addressed.

A similar prospective, multicenter comparative effectiveness study (the 4M study) included 626 patients who were biopsy-naïve at four centers in the Netherlands.[227] All patients underwent pre-biopsy MRI followed by systematic biopsy. Those with abnormal MRI also underwent in-bore MRI-guided biopsy. All MRIs were centrally reviewed by two highly experienced radiologists who were experts in prostate MRI. Overall, 49% of MRIs were read as normal (PI-RADS 1–2) and only 6% as indeterminate (PI-RADS 3). Clinically significant cancer (Grade Group ≥2) was detected in 30% of participants using the combined approach, in 25% using the MRI-targeted approach, and in 23% using a systematic approach. No difference in clinically significant prostate cancer detection was seen between targeted and systematic biopsy (difference, 2%; 95% CI, -1 to 5). Similar to PRECISION and MRI-FIRST, detection of insignificant cancer was lower using the targeted approach (difference, 11%; 95% CI, 7–14). Not biopsying those with PI-RADS 1–2 MRI missed only 4% of those with

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clinically significant prostate cancer detected on systematic biopsy. Among the 317 individuals with an abnormal MRI, 21 (7%) had cancer detected only on systematic biopsy. This study is remarkable for the high proportion of individuals (49%) who could have avoided biopsy based on a normal MRI while still maintaining an equivalent detection of clinically significant prostate cancer when compared to systematic biopsy for all individuals. Like MRI-FIRST and multiple retrospective studies, this study also shows that a relatively small proportion of clinically significant prostate cancer would be missed by omitting systematic biopsy. The authors acknowledge that this study represents the best-case scenario where MRI is performed and interpreted by experts and targeted biopsy is performed by experts. The results may not be widely generalizable without extensive training of radiologists and urologists, but the potential for the MRI-targeted approach is relatively high for achieving the goals of reducing biopsies, maximizing detection of clinically significant prostate cancer, and reducing overdetection of indolent cancer.

Another similar study, STHLM3-MRI, included 1532 individuals in Sweden with PSA levels ≥3 ng/mL who were randomly assigned 2:3 to undergo standard biopsy (n = 603) or to an experimental group that had an MRI evaluation with both targeted and standard biopsy if the MRI suggested the presence of prostate cancer.[228] As in the previously described trials, the experimental approach led to the diagnosis of fewer clinically insignificant cancers. It was also noninferior to standard biopsy for detecting clinically significant cancer.

In the Trio study, 2103 patients with prostate lesions visible by MRI underwent both MRI-targeted and systematic biopsy.[229] The primary outcome measure was cancer detection according to Grade Group. The rate of detection of Grade Group 1 cancers was significantly lower by MRI-targeted biopsy than by systematic biopsy, whereas the rate of detection of Grade Group ≥3 cancer was significantly higher with MRI-targeted biopsy ($P < .01$ for all comparisons). Of the 404 participants who underwent subsequent radical prostatectomy, upgrading to Grade Group ≥3 cancer occurred in 16.8% based on systematic biopsy, 8.7% based on MRI-targeted biopsy, and 3.5% based on the combination. The combined approach also resulted in 59 additional cancer diagnoses of Grade Group ≥3 than would have been found by use of either method alone.

The Canadian multicenter, prospective, randomized PRECISE clinical trial included 453 patients who were biopsy-naïve with a clinical suspicion of prostate cancer and PSA ≤20 ng/mL.[230] Participants were randomized to undergo MRI followed by targeted biopsy only if the PI-RADS score was ≥3 (MRI-TB) or systematic TRUS biopsy. The primary outcome measure, the proportion of participants with a diagnosis of Grade Group ≥2 cancer, was similar in the two groups (35% vs 30%, respectively; absolute difference, 5%; 97.5% 1-sided CI, -3.4% to infinity; noninferiority margin, -5%). Thus, the MRI-TB approach was noninferior to the systematic TRUS biopsy approach. Importantly, 37% of participants in the MRI-TB group had a negative MRI and avoided biopsy, adverse events were less common in the MRI-TB group, and the detection of Grade Group 1 cancer was lower in the MRI-TB group (22% vs 10%; risk difference, -11.6%; 95% CI, -18.2% to -4.9%).

The large, population-based, randomized GÖTEBORG-2 trial included >37,000 individuals aged 50 to 60 years who had regular PSA testing.[231] Participants with PSA levels ≥3 ng/mL received a prostate MRI and were randomized 2:1 to receive MRI-targeted biopsy only (experimental group) or targeted biopsy and systematic biopsy (reference group). All of the 5994 patients in the reference group were biopsied, whereas 262 of 11,986 participants (of 268 with positive MRI) underwent targeted biopsy in the experimental group. Results indicated that 66 individuals in the experimental group (0.6%) and 72 in the reference group (1.2%) received a diagnosis of clinically insignificant prostate cancer (95% CI, −1.0 to −0.4;

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RR, 0.46; 95% CI, 0.33–0.64; *P* < .001). Clinically significant prostate cancer was diagnosed in 110 individuals in the experimental group (0.9%) and 68 individuals in the reference group (1.1%; RR, 0.81; 95% CI, 0.60–1.10). Ten clinically significant prostate cancers, all of which were intermediate risk, were detected by systematic biopsy (0.2%). After additional rounds of screening and a median follow-up of 3.9 years, a recent analysis of the GÖTEBORG-2 trial found that more high-risk cancers were detected in the reference group (23) compared to the experimental (15), with an RR of 0.65 (95% CI, 0.34–1.24).[232] Data from this study thus show an approximately 50% decreased risk of detecting clinically insignificant prostate cancers when systematic biopsies are omitted, with diagnoses of intermediate-risk and high-risk cancers remaining about the same. However, the low biopsy rate in the experimental group makes it impossible to know the true numbers of missed cancers in that group, and the Panel continues to recommend the use of both systematic and targeted sampling when performing biopsies.

Many other studies support the notion that incorporating MRI-targeted biopsies into initial biopsies can improve the detection of clinically significant disease and also note that fewer significant upgrades are seen at radical prostatectomy in those who underwent initial biopsy using MRI targeting.[213,233-237] However, evidence from other studies is mixed as to whether MRI-guided biopsy improves clinically significant prostate cancer detection in individuals without a prior biopsy.[215,226,238] For example, a single-center trial randomized 130 individuals who were biopsy-naïve to a control group that received TRUS-guided random biopsy alone or to a group that received pre-biopsy mpMRI, TRUS-guided random biopsy, and cognitive MRI/TRUS fusion-targeted biopsy.[215] Similar rates of detection of prostate cancer (64% vs 57%; *P* = .5) and of clinically significant cancer (55% vs 45%; *P* = .8) were seen in the two arms. In another randomized trial, 212 patients who were biopsy-naïve with suspected prostate cancer were assigned to a pre-biopsy mpMRI group or to a standard biopsy group.[238] Participants in the mpMRI group had targeted fusion biopsies if suspicious lesions were seen. Otherwise, they received standard biopsies. More clinically significant prostate cancers were detected in the mpMRI arm (43.9% vs18.1%; *P* < .001).

Similar to the studies above, multiple retrospective studies have shown that adding MRI-targeted biopsy to systematic biopsy increases the yield of clinically significant prostate cancer over systematic biopsy alone. A 2019 Cochrane systematic review identified 18 cross-sectional studies that compared template-guided biopsy with MRI only, MRI-targeted biopsy, MRI with or without MRI-targeted biopsy, and/or systematic biopsy for the detection of Grade Group ≥2 prostate cancer.[239] The authors concluded that the use of MRI to guide biopsy decisions, with or without MRI-targeted biopsy, led to fewer missed Grade Group ≥2 cancers and a decrease in the detection of Grade Group 1 cancers compared with systematic biopsy alone. Results of another systematic review suggest that the approach of MRI first followed by targeted biopsies (with or without systematic biopsies) in individuals who are biopsy-naïve may improve detection of clinically significant cancer, reduce the number of biopsy cores per procedure, reduce adverse effects, and potentially reduce unnecessary biopsies compared with systematic biopsies alone.[240]

In summary, the data suggest that a significant proportion of people who are biopsy-naïve with elevated PSA can avoid biopsy based on a PI-RADS ≤2 result on mpMRI (21%–49%), but some of these people have clinically significant disease that would be missed (2%–14% of the study populations).[220,226,227,230,241-243] It is important to note that a similar or possibly higher proportion of clinically significant disease is missed with a traditional approach of proceeding to a systematic biopsy without a mpMRI, with higher detection of indolent disease.[226,244]

The Panel also notes that the vast majority of published evidence using MRI for prostate cancer diagnosis comes from high-volume centers of

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excellence. The generalizability of these findings is not yet clear. In light of evidence showing considerable interobserver variability in the interpretation of prostate MRI, the Panel emphasizes the need for high-quality mpMRI and radiologic expertise for optimal reading of scans.[245-247] The PI-RADS from the American College of Radiology gives recommendations for high-quality MRI in prostate cancer care, including recommendations related to the use of MRI to direct targeted biopsies (https://www.acr.org/Clinical-Resources/Clinical-Tools-and-Reference/Reporting-and-Data-Systems/PI-RADS).

At this time, the Panel strongly recommends that an mpMRI (category 1) evaluation be performed before proceeding to an image-guided biopsy to inform biopsy decisions and to help identify regions of the prostate that may harbor cancer. The Panel notes that the availability of mpMRI has grown in recent years, such that more than a third of patients received MRI before biopsy in 2022.[248]

The Panel cautions that false negatives can occur with mpMRI, even with the best equipment and readers due to the negative predictive value (NPV) for clinically significant prostate cancer (around 86%–98%).[220,226,227,230,241,242] Therefore, proceeding to image-guided biopsy following a negative MRI should still be considered, particularly in situations where the patient is considered to be at high risk for cancer based on risk calculators, PSA density, or other biomarkers.[199,222,249,250]

The Panel also recommends that image-guided biopsy techniques should be used routinely (see *Targeted Biopsy Techniques*, below). Radiologic expertise and the use of high-quality mpMRI hardware is essential for optimal interpretation of scans. Although some advocate for excluding systematic biopsy in those undergoing MRI targeting (due to concerns that it may increase the risk of overdiagnosis), most advocate for a combined approach (preferred) as some clinically significant prostate cancers are uniquely detected using the systematic approach and systematic biopsies are needed for risk stratification if cancer is found.

In individuals with at least one negative biopsy, the Panel believes that high-quality mpMRI may help identify regions of cancer missed on prior biopsies; it should therefore be considered in this setting if not done previously (also see *Repeat Biopsies*, below).[251]

Biparametric MRI

As discussed above, the use of mpMRI is a preferred (category 1) diagnostic tool for accurately detecting and localizing prostate cancer. However, due to various circumstances, mpMRI, which includes dynamic contrast-enhanced (DCE) imaging, is not always available to individuals who require MRI for prostate cancer identification and staging. Biparametric MRI (bp-MRI), which does not include DCE imaging, offers a diagnostic alternative to patients and clinicians that allows measurement of both the prostate and lesion volume. The use of bp-MRI can shorten scan times, meet increasing demand for diagnostic workups, lower costs, increase patient comfort, and reduce the risk of allergic reactions to imaging agents such as gadolinium while remaining diagnostically accurate for clinically significant prostate cancer detection. In one study, several diagnostic values of bp-MRI and mpMRI were equal, including sensitivity of 95% (95% CI, 91–97), specificity of 69% (95% CI, 64–73), NPV of 97% (95% CI, 94–98), and PPV of 57% (95% CI, 51–62).[252] A further study reached similar conclusions when analyzing data from the PROMIS trial.[253] The authors reported values of sensitivity (94% and 95%), specificity (37% and 38%), PPV (51% and 51%), and NPV (90% and 91%) for MRI without and with DCE imaging ($P > .05$) in cases of significant prostate cancer. A systematic review and meta-analysis published in 2025 also concluded that bp-MRI and mpMRI have comparable sensitivity and specificity.[254] It is noted that among expert

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diagnostic readers, there is some uncertainty that may arise without dynamic contrast.[255]

The prospective, multi-center PRIME trial tested the noninferiority of bp-MRI compared to mpMRI for the detection of clinically significant prostate cancer in individuals who were biopsy-naïve.[256] PRIME included 490 participants with elevated PSA levels, who underwent mpMRI. The bp-MRI results were reported first, while the radiologists were blinded to the DCE sequence. Full mpMRI results were reported after unblinding. Targeted biopsies, with or without systematic biopsies, were performed for all patients with bp-MRI or mpMRI results that were concerning for clinically significant cancer. The results indicated that bp-MRI met the noninferiority margin of 5%, with clinically significant prostate cancer detected in 29.2% of individuals compared with 29.6% using the mpMRI results (difference, −0.4; 95% CI, −1.2 to 0.4; $P = .50$). The Panel notes that this trial was performed at highly experienced centers with experienced readers and that whether these results can be applied at less experienced centers remains uncertain. Therefore, the Panel continues to recommend mpMRI as the preferred option (category 1).

Micro-Ultrasound

Micro-US is technique that produces images at approximately 300% higher resolution than conventional US, allowing for visualization of prostatic ducts and fine architectural details.[257] Micro-US can reach a depth of 5 to 6 cm, which may limit the detection of prostate cancer outside the peripheral zone in very large prostates.[257] It has a sensitivity of 80% and a specificity of 37% for the detection of clinically significant prostate cancer in the peripheral zone.

Micro-US may be more accessible and less expensive than mpMRI and does not require the use of a contrast agent. Furthermore, at experienced centers, micro-US has been shown to offer similar diagnostic quality as mpMRI. Therefore, like bp-MRI (discussed above), micro-US can be used as an alternative option in patients with an elevated PSA when mpMRI is not available. The Panel notes that training and expertise are critical for use of micro-US in this setting and that mpMRI continues to be preferred.

The OPTIMUM trial assessed the use of micro-US for biopsy targeting.[258] OPTIMUM was a multicenter, international, open-label, randomized, non-inferiority trial that randomized 802 biopsy-naïve patients with elevated PSA. Of those patients, 678 underwent biopsy: micro-US-guided biopsy (n = 121), micro-US/MRI fusion-guided biopsy (n = 226), or MRI/conventional US fusion-guided biopsy (n = 331). All participants also underwent a systematic biopsy. The primary outcome was the difference in detection of Grade group ≥2 prostate cancer between the first and third groups described above, with a noninferiority margin of 10%. This measure was met with a 3.52% higher rate of diagnosis in the micro-US group (95% CI, −3.95% to 10.92%; noninferiority $P < .001$). Micro-US/MRI fusion was also noninferior to MRI/conventional US fusion, the trial's secondary outcome (difference, 4.29%; 95% CI, −4.06% to 12.63%; noninferiority $P < .001$). There was no significant difference in the detection of Grade group 1 cancers between the groups.

DRE

DRE should not be used as a stand-alone method for the early detection of prostate cancer, because DRE alone misses a substantial number of clinically significant cancers.[259,260] In addition, prostate cancers detected solely by DRE are less likely to be confined to the prostate than those detected through PSA testing, and the positive predictive value (PPV) for DRE alone is poor.[191,261-264]

DRE Alongside PSA Testing

The use of DRE as part of a PSA-based early detection program is controversial. Although the absolute numbers may be small, some

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clinically significant cancers may be missed using a serum PSA cut point alone; it is estimated that around 0.2% of all individuals screened may have DRE-detectable prostate cancer with PSA <3 ng/mL.[265,266] Furthermore, cancer detection rates may be higher when DRE is used with PSA than when PSA is used alone, and patients with an abnormal DRE and an elevated PSA may be more likely to be diagnosed with Grade Group ≥2 prostate cancer than those with only an elevated PSA.[207,263,267]

However, the evidence described above comes from an era prior to the current MRI-directed pathway for prostate cancer diagnosis, and there is little evidence regarding the added value of DRE if an MRI will be obtained regardless of DRE findings. Furthermore, a quality DRE requires assessment of nodularity, induration, fixation, and asymmetry. Inter-examiner variability is a known issue. Not all clinicians are trained to perform a DRE as well as those conducting the above-referenced trials, suggesting that the benefit in a community physician's office may be lower than in the cited studies. Although there is little to no risk associated with a DRE, the procedure can be uncomfortable for patients and could therefore present a barrier to prostate cancer early detection.

Thus, DRE should be considered optional as part of PSA-based early detection. Patients opting for an early detection program should be offered PSA with an option for the addition of DRE. Individuals with a DRE that is very suspicious for cancer should undergo further evaluation regardless of PSA results. However, the Panel notes that, since the PPV of a DRE in those with a normal PSA is poor, an abnormal DRE result alone as an indication for biopsy would lead to a large number of unnecessary biopsies and the detection of many insignificant cancers in individuals with low PSA values.[191,263,264,268]

DRE in Patients with Elevated PSA

Evidence suggests that the utility of DRE may be greatest in those with an elevated PSA. An analysis of the PLCO trial found that a suspicious DRE was associated with the identification of Grade Group ≥2 prostate cancer in those with a PSA ≥3 ng/mL (23.0% risk at 10 years vs 13.7% risk at 10 years in those with a non-suspicious DRE), but not in those with a PSA <2 ng/mL (1.5% vs 0.7%).[269] Ten-year risk in those with a PSA in the 2 to 3 ng/mL range were 6.5% in individuals with a suspicious DRE compared with 3.5% in individuals with a non-suspicious DRE. Another study also showed that among those with a serum PSA >3 ng/mL, those with a positive DRE were more likely to have prostate cancer.[193]

Thus, DRE should be performed for anyone with an elevated PSA to aid in decisions regarding biopsy (see *Pre-Biopsy Workup*, above).

Biomarker Testing

Several biomarker tests have been developed with the goals of refining which patients should consider biopsies, decreasing unnecessary biopsies, and increasing the specificity of cancer detection, without missing a substantial number of higher-grade (Grade group ≥2) cancers. These tests have been studied in patients who were biopsy-naïve and in patients who have had one negative biopsy to determine if repeat biopsy is an appropriate consideration.

The extent of validation of these tests varies across diverse populations and is mostly limited to retrospective observational studies, often funded with industry support (see below). Results of biomarker assays can be complex and should be interpreted with caution. It is important for clinicians and patients to understand that there are risks in using biomarker tests that lack robust validation to change management decisions, as they may drive patients or providers to inappropriate options.

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In particular, these tests may underestimate risk, leading to delayed or missed diagnoses.

Furthermore, the Panel emphasizes that these tests should not be used in place of mpMRI. When a patient meets the standards for biopsy, mpMRI is recommended based on high-level data (category 1). If mpMRI is not available, contraindicated, or of low-quality or if the patient and/or the physician wish to more precisely estimate risk of Grade Group ≥2 prostate cancer compared to PSA, certain biomarker tests that meet criteria of both validity and clinical utility as outlined in the guidelines above can be considered before proceeding to immediate biopsy (category 2B).

Head-to-head comparisons have been performed for some of these tests, used independently or in combinations in the initial or repeat biopsy settings, but sample sizes were often small and results varied.[270-282] Therefore, the Panel believes that no biomarker test can be recommended over any other at this time.

Results of any of these tests, when performed, should be included in discussions between the clinician and patient to assist in decisions regarding whether to proceed with biopsy.

%fPSA

Unbound or free PSA (fPSA), expressed as a ratio of tPSA, is a clinically useful molecular form of PSA, with the potential to improve early detection, staging, and monitoring of prostate cancer. Several molecular forms of PSA are known to circulate in the blood. In most individuals, the majority (60%–90%) of circulating PSA is covalently bound to endogenous protease inhibitors. Most immunoreactive PSA is bound to the protease inhibitor alpha-1-antichymotrypsin. Other immunoreactive PSA-protease inhibitor complexes, such as alpha-1-antitrypsin and protease C inhibitor, exist at such low serum concentrations that their clinical significance has not been determined. In addition, a large proportion of PSA is complexed with alpha-2-macroglobulin (AMG). Unfortunately, this PSA-AMG complex cannot be measured by conventional assays because of the shielding (or "caging") of PSA antigenic epitopes by AMG.

Most clinical work investigating the use of the molecular forms of PSA for early detection of prostate cancer has focused on the percentage of PSA found circulating in the free or unbound form. Numerous studies have shown that the %fPSA is significantly lower in patients who have prostate cancer compared with those who do not.

The U.S. Food and Drug Administration (FDA) approved the use of %fPSA for the early detection of prostate cancer in individuals aged ≥50 years with a non-suspicious DRE and PSA levels between 4 and 10 ng/mL (PSA levels where most secondary testing is done). The multi-institutional study that characterized the clinical utility of this assay showed that a 25% fPSA cutoff detected 95% of prostate cancers while avoiding 20% of unnecessary prostate biopsies.[283] A more recent study showed that a cutoff of 17.6% provides a sensitivity of 86.5%, specificity of 63.7% in patients with PIRADS<3 and PSA ≤10 ng/mL.[284] Furthermore, an analysis of participants in the PLCO screening trial showed that using %fPSA with PSA improved the prediction of both clinically significant and fatal prostate cancer.[285]

%fPSA is included in the guidelines as an option for individuals with a prior negative biopsy.

PHI

The Prostate Health Index (PHI) is a combination of the tPSA, fPSA, and proPSA tests.[286-288] In a multicenter study, it was noted to have approximately double the sensitivity of %fPSA for cancer detection in those with serum PSA concentrations between 2 and 10 ng/mL.[289] In addition, the PHI correlated with cancer grade and had an AUC of 0.72 for discrimination of Grade Group ≥2 cancer from low-grade cancer or

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negative biopsy. Another prospective cohort study calculated an AUC of 0.815 for the detection of Grade Group ≥2 prostate cancer.[290] This study determined the optimal cutoff of PHI to be a score of 24, which should lead to 36% of biopsies avoided with approximately 2.5% of high-grade cancers missed. Other studies have also shown that PHI can predict aggressive prostate cancer and has potential clinical utility.[291-294]

The PHI was approved by the FDA in 2012 for use in those with serum PSA values between 4 and 10 ng/mL. A clinical utility study conducted at four large urology group practices showed that use of PHI was in fact associated with a decrease in biopsy procedures performed when compared to historical controls from the same physicians (36.4% vs 60.3%; $P < .0001$).[295] Patients in the study had normal DRE and PSA values ranging from 4 to 10 ng/mL. Physician survey results showed that PHI results impacted biopsy decisions for 73% of patients. However, the authors of this study did not report the numbers of Grade Group ≥2 cancers missed, and some have estimated that it may be as high as 30%.[296]

PHI is included as an option in the guidelines in both the pre- and post-biopsy settings.

4Kscore

The 4Kscore test is another combination test that measures fPSA, tPSA, human kallikrein 2 (hK2), and intact PSA and also considers age, DRE results, and prior biopsy status.[297,298] This test reports the percent likelihood of finding Grade Group ≥2 cancer on biopsy. Multiple retrospective studies of large cohorts have shown high discrimination between those with Grade Group ≥2 tumors from those with benign or Grade Group 1 tumors (AUC typically >0.80).[299-302] These results suggest a large proportion of biopsies (>40%) could potentially be avoided, at the cost of missing a smaller percentage (10%–20%) of Grade Group ≥2 cancers. These studies used different cutoffs for which patients should have a biopsy, making it difficult to assess how much the results can be generalized.

A multicenter clinical utility study found a 65% reduction in prostate biopsies with use of the 4Kscore test.[303] In addition, a correlation between 4Kscore risk category and Gleason score was seen ($P < .01$). Data from 205 Black patients in a single Veteran Affairs institution showed the ability of 4Kscore to predict Grade Group ≥2 prostate cancer in Black patients can be improved using a race coefficient.[304]

The 4Kscore test was approved by the FDA in December 2021 to aid in detection of aggressive prostate cancer in patients recommended for a biopsy. The Panel consensus is that the test can be considered for patients prior to biopsy and for those with prior negative biopsy who are thought to be at higher risk for clinically significant prostate cancer. It is important for patients and their urologists to understand, however, that no optimal cut-off threshold has been established for the 4Kscore. If a 4Kscore test is performed, patients and their urologists should discuss the results to decide whether to proceed with a biopsy.

ConfirmMDx

ConfirmMDx is a tissue-based, multiplex epigenetic assay that aims to improve the stratification of individuals being considered for repeat prostate biopsy. Hypermethylation of the promoter regions of *GSTP1, APC,* and *RASSF1* is assessed in core biopsy tissue samples, normalized to give an epigenetic score, sometimes called EpiScore. The test is performed in one Clinical Laboratory Improvement Amendments (CLIA)-certified laboratory.

The European MATLOC study blindly tested this assay in archived tissue from 498 patients with negative biopsies who had repeat biopsies within 30 months.[305] The NPV was 90% (95% CI, 87%–93%). In a multivariate

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analysis, ConfirmMDx was predictive of patient outcomes (OR, 3.17; 95% CI, 1.81–5.53). A similar validation study was performed in the United States using archived tissue from 350 patients with negative biopsies who had repeat biopsies within 24 months.[306] Using the same thresholds as the MATLOC study, the NPV was 88% (95% CI, 85%–91%), and the test was again found to be predictive of outcomes on multivariate analysis (OR, 2.69; 95% CI, 1.60–4.51). Additional studies have shown a correlation between the epigenetic score and Grade Group ≥2 prostate cancer.[307,308] Furthermore, a study in 211 African American individuals found that the test worked similarly as it did in previous studies, which predominantly included white individuals.[309]

The Panel believes that ConfirmMDx can be considered as an option for individuals contemplating repeat biopsy, because the assay may identify individuals at higher risk of prostate cancer diagnosis on repeat biopsy. This assay is approved for limited coverage by MolDX for the reduction of unnecessary repeat prostate biopsies.

ExoDx Prostate (IntelliScore)

ExoDx Prostate (IntelliScore), also called EPI, evaluates a urine-based 3-gene exosome expression assay utilizing *PCA3* and *ERG* RNA from urine, normalized to *SPDEF*. The background for these markers is supported by a number of studies, but the application to exosome detection is unique.[310] This gene panel proposes to discriminate Grade Group ≥2 prostate cancer from Grade Group 1 and benign disease at initial biopsy. The population for which use of the assay was initially intended to include patients >50 years with no prior biopsy and a PSA value between 2 and 10 ng/mL. In the initial study, the estimate of the AUC for the assay was 0.71 in the validation cohort, with significant improvements when the test was added to standard-of-care variables compared with standard variables alone.[311] Use of the test decreased the need for biopsy by 27% (138 of 519) while missing 8% (12 of 148) of Grade Group ≥2 cancers. The majority of exclusions were for urine volume >49 mL, assay failure, and application outside the intended use population.

A second independent validation study was a 2-phase adaptive clinical utility study that included 503 patients who were biopsy-naïve with PSA levels between 2 and 10 ng/mL and compared EPI and biopsy results.[312] In the first phase of this study, the AUC was 0.70 for predicting Grade Group ≥2 cancer by EPI. Using the validated cut point of 15.6, the test has an NPV of 89%, reducing total biopsies by 20% and missing 7% of Grade Group ≥2 cancer. The second phase of this trial will be reported in the future.

In a clinical utility study, 1094 patients (with 72 urologists from 24 urology practices) were randomized to receive EPI results before biopsy or not. The EPI test results influenced the decision to proceed to biopsy in many instances, and 30% more Grade Group ≥2 prostate cancers were diagnosed than in the control arm.[313]

Interim results of a randomized, blinded clinical utility trial have been published.[314] The 1049 participants had PSA levels of 2 to 10 ng/mL and were being considered for prostate biopsy. All participants had an EPI test, but only those randomized to the EPI group received the results before making a decision regarding biopsy. Follow-up data was available for 833 participants at 2.5 years. Biopsy rates were lower for low-risk EPI scores than high-risk EPI scores in the EPI group (44.6% vs 79.0%; $P < .001$), but not in the control group (59.6% vs 58.8%; $P = 0.99$). However, only 21.2% of participants had a low EPI. Overall, the EPI group saw a 13% absolute increase in biopsy rate (72.0% in the EPI group compared to 59.0% in the control group). These additional biopsies led to a small absolute increase in Grade Group ≥2 cancers diagnosed (24.8% vs 20.6% in the EPI and control groups, respectively). However, use of EPI also led to a similar absolute increase in Grade Group 1 diagnoses (15.1% vs 10.0% in the EPI and control groups, respectively).

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The use of EPI in the post-initial biopsy setting was studied as part of a prospective validation study that included 229 patients.[315] Using the previously validated cutoff, the NPV was 92%, with 26% of unnecessary biopsies avoided and 2.1% of Grade Group ≥2 prostate cancers missed.

The Panel believes that EPI can be considered as an option for individuals contemplating initial or repeat biopsy based on the high NPV, though caution is warranted, as routine use of EPI may increase overall biopsy rates and overdiagnosis of low-grade cancer.

SelectMDx

SelectMDx is a gene expression assay performed on post-DRE urine that measures *DLX1* and *HOXC6* expression against *KLK3* as internal reference. *DLX1* and *HOXC6* have been associated with prostate cancer aggressiveness.[316,317] As with other assays, SelectMDx is designed to improve the identification of individuals with Grade Group ≥2 prostate cancer prior to biopsy, thereby reducing the number of unnecessary biopsies.

The assay was developed on an initial training set of 519 patients from two prospective multicenter studies and was then validated in a separate set of 386 patients from these trials.[318] Using the expression of *DLX1* and *HOXC6* alone resulted in an AUC of 0.76, a sensitivity of 91%, a specificity of 36%, an NPV of 94%, and a PPV of 27% for the prediction of Grade Group ≥2 prostate cancer.

A multicenter study used pre-biopsy urine samples from 1955 individuals to validate the assay with a training cohort and a validation cohort.[319] The AUC was 0.85, the sensitivity was 93%, the specificity was 47%, and the NPV was 95% for detection of Grade Group >2 prostate cancer in the 916-patient validation cohort. When only those with PSA levels <10 ng/mL were included, the values were 0.82, 89%, 53%, and 95%, respectively.

The use of this assay with or without mpMRI was assessed in a prospective study that included 599 biopsy-naïve participants with PSA levels ≥3 ng/mL who all received the test and an mpMRI.[320] Simulation of different strategies showed that mpMRI had the most clinical benefit (avoiding biopsy in 49% of participants and missing 4.9% of Grade Group ≥2 prostate cancer). Limiting mpMRI to those with low-risk by SelectMDx avoided more biopsies (60%) but missed 13% of Grade Group ≥2 cancers. The authors concluded that SelectMDx could be helpful in cases where mpMRI is not available (avoiding biopsy in 38% and missing 10% of Grade Group ≥2 cancers). Another prospective study found similar results.[321] The authors of this study noted that using SelectMDx after a negative mpMRI to determine who might still need a biopsy led to fewer missed Grade Group ≥2 cancers but also led to more unnecessary biopsies. Performing biopsies only on patients with a PI-RADS 3-5 lesion would avoid 57% of biopsies and miss 16% of Grade Group ≥2 cancers, whereas adding biopsies for individuals with a positive SelectMDx test and a negative mpMRI would avoid 40% of biopsies and miss 3% of Grade Group ≥2 prostate cancers.

Overall, the Panel believes that SelectMDx score is potentially informative in patients who have never undergone biopsy, and it can therefore be considered in such individuals.

IsoPSA

IsoPSA is a blood-based assay that uses an aqueous 2-phase system to partition and detect PSA isoforms in blood.[322] In a prospective, multicenter validation study, IsoPSA was assessed as a diagnostic biomarker test in 888 patients scheduled for initial or repeat prostate biopsy.[323] IsoPSA demonstrated an AUC of 0.78 (95% CI, 0.75–0.81) for Grade Group ≥2 prostate cancer, with a sensitivity of 90.2% (95% CI, 86.4–93.0), specificity of 45.5% (95% CI, 41.4–49.6), PPV of 47.7% (95% CI, 45.7–49.8), and NPV of 89.3 (95% CI, 85.6–92.2).

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In a real-world clinical utility study (n = 900), 38 clinicians practicing across a range of academic and community settings altered their pre-test recommendations for or against biopsy and/or MRI for 66% of patients after using the IsoPSA test in clinical practice, resulting in a net decrease in biopsy recommendations of 55%.[324]

Results of a single-center retrospective review of patients with PSA >4 ng/mL who had received an IsoPSA test and a prostate MRI suggest that patients with PI-RADS 1–3 and a low IsoPSA ratio have a <5% probability of having Grade Group ≥2 prostate cancer.[325]

The Panel believes that IsoPSA can be considered for patients prior to biopsy and for those with prior negative biopsy who are thought to be at higher risk for clinically significant prostate cancer.

MyProstateScore

The MyProstateScore (MPS) assay measures total serum PSA and post-DRE urine expression of *PCA3* and the *TMPRSS2::ERG* fusion gene.[326] Rearrangements of the *ERG* gene are found in approximately half of prostate cancers.[327] The *TMPRSS2::ERG* fusion specifically occurs at high frequency and appears to be an early event in prostate cancer development.[328] The role of *PCA3* in prostate cancer is discussed above. Early studies suggested that the combination of these two markers improved the prediction of prostate cancer on biopsy.[329]

An MPS validation study included 1244 individuals with planned biopsy (80% with no prior prostate biopsy) in a validation cohort.[326] The AUC for the prediction of any cancer was 0.75 for MPS, compared with 0.59 for PSA alone. For the prediction of Grade Group ≥2 cancer, the AUCs for MPS and PSA alone were 0.77 and 0.65, respectively.

Another validation cohort included 1525 patients from academic and community centers who were referred for initial biopsy.[330] MPS testing provided a sensitivity of 97% and an NPV of 98% for Grade Group ≥2 prostate cancer. The authors calculate that 33% of unnecessary biopsies could have been avoided by using the assay in biopsy decisions, while missing 3% of Grade Group ≥2 cancers.

The Panel believes that MPS can be considered for patients prior to biopsy and for those with prior negative biopsy who are thought to be at higher risk for clinically significant prostate cancer.

MyProstateScore 2.0

In order to improve diagnostic accuracy for high-grade prostate cancer in individuals with elevated PSA levels, the MPS 2.0 assay (MPS2) was developed to measure 18 markers that are overexpressed in Grade Group ≥2 compared to Grade Group 1 prostate cancer.[331] Researchers analyzed post-DRE urine samples from 761 individuals with elevated PSA levels prebiopsy, and reported an AUC value for Grade Group ≥2 cancer of 0.81 (95% CI, 0.77–0.85) for MPS2, 0.82 (95% CI, 0.78–0.86) for MPS2+ (which accounts for prostate volume), and 0.74 (95% CI, 0.69–0.78) for MPS. External validation in 743 individuals revealed superior diagnostic accuracy for Grade Group ≥2 prostate cancers with MPS2 compared to certain other available options. The sensitivity, specificity, NPV, and PPV of MPS2 were 95%, 37%, 97%, and 28%, respectively. The authors estimated that 370 unnecessary biopsies could be avoided per 1000 individuals.

A validation study of MPS2 testing in urine specimens from individuals who did not undergo DRE concluded that MPS2 showed noteworthy value for detection of Grade Group ≥2 prostate cancer when measured against PSA-based tests.[332] MPS2 testing led to superior diagnostic accuracy with an AUC of 0.71 (95% CI, 0.65–0.77) compared to PSA alone, which had an AUC of 0.57 (95% CI, 0.50–0.64), and PCPT risk calculator with AUC of 0.62 (95% CI, 0.55–0.69). In this non-DRE population, the sensitivity,

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specificity, NPV, and PPV of MPS2 were 92%, 35%, 93%, and 32% respectively in the pre-biopsy setting and 93%, 44%, 95%, and 36%, respectively in the post-biopsy setting. These values were increased when incorporating biomarkers, clinical factors, and prostate volume model in MPS2 testing.

The Panel believes that MPS2 can be considered for patients prior to biopsy and for those with prior negative biopsy who are thought to be at higher risk for clinically significant prostate cancer.

Stockholm3

Stockholm3 is a multivariate model that includes clinical variables (ie, age, first-degree family history of prostate cancer, previous biopsy), blood biomarkers (ie, PSA, fPSA, %fPSA, hK2, MIC1, MSMB), a genetic score based on 254 single-nucleotide polymorphisms (SNPs) and an explicit variable for the HOXB13 SNP, prostate volume, and DRE results.[333] When used in Swedish patients with PSA ≥3 ng/mL, the test reduced the number of biopsies by 34% compared with PSA alone. Sensitivity was similar between the multivariate model and PSA alone.

In a prospective, population-based, randomized, open-label, noninferiority trial in Sweden (STHLM3-MRI), 2293 participants with either a PSA ≥3 ng/mL or a Stockholm3 score above the set threshold were randomly assigned 2:3 to either systematic prostate biopsies or to bp-MRI followed by MRI-targeted and systematic biopsy in those with positive MRIs.[334] The AUC for the detection of Grade Group ≥2 prostate cancer was 0.76 (95% CI, 0.72–0.80) for Stockholm3 and 0.60 (95% CI, 0.54–0.65) for PSA. Screening with Stockholm3 and the MRI approach was associated with higher detection of Grade Group ≥2 prostate cancer, lower detection of low-grade cancers, and led to fewer biopsy procedures than screening using the PSA and systematic biopsy approach. External validation in a Germon cohort found that Stockholm3 could reduce 52% of biopsies while missing 8% of Grade Group ≥2 cancers.[335]

Similar conclusions were reached in a prospective, multicenter validation study involving 342 undiagnosed individuals scheduled for prostate biopsy. The report concluded that blood-based Stockholm3 testing demonstrated superior distinction (AUC, 0.77; 95% CI, 0.72–0.82) of Grade Group ≥2 prostate cancer compared to PSA with an AUC of 0.66 (95% CI, 0.61–0.72).[336] Notably, the ≥0.11 cutoff score of Stockholm3 found 92% of Grade Group ≥2 prostate cancers, 23% higher than the PSA ≥1.5 ng/mL cutoff (69%). A separate prospective observational study, which included a diverse, multi-ethnic cohort, showed that the sensitivity of Stockholm3 was noninferior to PSA ≥4 ng/mL (relative sensitivity, 0.95; 95% CI, 0.92–0.99) and had a higher specificity (relative specificity, 2.91; 95% CI, 2.63–3.22).[337] The authors estimated that Stockholm3 could reduce unnecessary biopsies by 45%.

A randomized trial in Sweden compared an approach of systematic biopsies in those with a Stockholm3 score ≥0.15 (biomarker group) to an approach using MRI in those with PSA ≥3 ng/mL and both systematic and targeted biopsies in those with PI-RADS 3–5 (MRI-enhanced group).[338] Similarity in detection rates between these two methods would be advantageous for areas with limited MRI access. Of the 12,743 participants, 7609 were assigned to the MRI-enhanced group, and 5134 were assigned to the biomarker group. Both approaches had comparable detection rates for Grade Group ≥2 prostate cancer (2.3% vs 2.5%; relative proportion, 0.92; 95% CI, 0.73–1.15), though a 1.9% increase in biopsies (6.3% vs 4.4%; relative proportion, 1.43; 95% CI, 1.23–1.66) and more than twice as many indolent prostate cancers (1.2% vs 0.5%; relative proportion, 2.21; 95% CI, 1.49–3.27) were observed in the biomarker group.

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The use of the Stockholm3 test with MRI was also studied in a prospective cohort of 532 participants, who all had MRI and Stockholm3.[339] All participants had a systematic biopsy and those with PIRADS 3–5 lesions also underwent a targeted biopsy. With a Stockholm3 score cutoff of >10%, 32% to 38% of biopsies could be avoided while 5% to 11% of Grade Group ≥2 cancers would be missed. The authors concluded that Stockholm3 risk prediction models will need to be recalibrated for use in the MRI approach.

The Panel believes that Stockholm3 can be considered for patients who are biopsy-naïve and thought to be at higher risk for clinically significant prostate cancer.

Additional Biomarker Tests

The list of assays with the potential to permit improved detection of Grade Group ≥2 prostate cancers as an adjuvant to PSA screening is growing rapidly. The Panel wants to highlight that there are many potential sources of error in current approaches to evaluating biomarkers, including a rate of undetected cancers in patients with a single negative prostate biopsy as high as 25%. Other significant and unaddressed issues include the well-known upgrading (32%–49%) that occurs in patients with Grade Group 1 cancer at biopsy at the time of pathologic assessment of the surgical specimen. Longer-term follow-up of the cohorts to determine whether missed prostate cancers were ultimately detected is needed. In addition, validation of these tests in other cohorts is needed before they can be accepted as alternatives to (or perhaps preferable to) other tests, described above.

Biopsy Technique

Initial Biopsy

Image-guided biopsy with targeting (preferred) or without targeting of lesions seen on pre-biopsy MRI is the recommended technique for prostate biopsy. When systematic biopsy is performed, the Panel recommends an extended-pattern, at least 12-core biopsy (sextant medial and lateral peripheral zone and lesion-directed). This extended-pattern scheme has been validated and results in enhanced cancer detection compared to sextant biopsy schemes.[340,341]

Image-guided biopsy can be performed via a transrectal or a transperineal approach.[342-344] The randomized TRANSLATE trial found that ultrasound-guided transperineal biopsy detected more Grade Group ≥2 cancer than a transrectal approach in participants who were biopsy-naïve (60% vs 54%; odds ratio, 1.32; 95% CI, 1.03–1.70; $P = .03$).[345] In addition, transperineal biopsy was associated with a lower risk of infections requiring hospital admission within 35 days of biopsy (0.4% vs 1.6%; also see *Risks of Biopsy*, below). However, extensive perineal template biopsies may lead to higher rates of other complications such as urinary retention. Currently, the Panel views both approaches as reasonable options.

Targeted Biopsy Techniques

Interest in the use of novel imaging, particularly MRI, to guide needle placement during biopsy (see *Magnetic Resonance Imaging,* above) has been increasing over the last decade.

Targeted biopsy techniques include cognitive or visual targeting (guiding with ultrasound [US], based on an MRI image), TRUS-MRI fusion platforms (merging a stored MRI image with a real-time US image), and direct in-bore magnetic resonance (MR)-guided biopsy (performed by an interventional radiologist while the patient is in the scanner).[346-348] Data show that mpMRI followed by lesion targeting increases the detection of clinically significant (Grade Group ≥3) disease while lowering the detection of low-risk (Grade Group 1) disease. Data also suggest that different targeting techniques detect clinically significant prostate cancer at similar rates.[349]

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Evidence from clinical trials and other studies evaluating MRI-targeted biopsy in the initial biopsy setting is described above (see *Magnetic Resonance Imaging*, above).

Overall, the Panel believes that the data for the use of MRI and MRI-targeted biopsies in the initial biopsy setting are increasingly compelling. However, studies using both targeted and systematic sampling routinely demonstrate higher yield of clinically significant cancer with the combined approach and improved sensitivity.[233,350] Therefore, a combination of systematic and targeted procedures is preferred when MRI-targeting capabilities are available, at least at initial biopsy.

Repeat Biopsies

A negative biopsy does not preclude a diagnosis of prostate cancer on subsequent biopsy. If clinical suspicion of cancer persists after a negative biopsy, consideration can be given to the use of mpMRI followed by an appropriate targeted biopsy technique based on the results. In addition, biomarker testing can also be considered in these individuals to inform decisions regarding repeat biopsy (category 2B; see *Biomarker Testing*, above).

Targeted Biopsy for Repeat Biopsy

After one or more negative biopsies, individuals who are considered at high risk (eg, those with persistently elevated or rising PSA) should be considered for MRI followed by targeted biopsy based on several studies showing improved detection of clinically significant prostate cancer in this setting.[211,351-357] Reported cancer detection rates by targeted fusion biopsies in individuals with previous negative biopsies range from 34% to 51%.[211,352-354] Studies that used direct MR guidance for targeted biopsies have reported similar cancer detection rates in those with previous negative biopsies: 41% to 56%.[355-357]

The targeted biopsy approach is associated with a higher rate of detection of clinically significant cancer in individuals with prior negative biopsy than repeat systematic biopsies, which lead to the identification of more low-risk tumors. For instance, in one retrospective cohort study, 105 participants with prior negative biopsies and elevated PSA underwent mpMRI followed by standard 12-core systematic biopsy and MR-US fusion-targeted biopsy regardless of MRI results.[353] Prostate cancer was found in 36 individuals (34%). In this study, 21 of 23 cancers (91%) identified by targeted biopsy were clinically significant (Grade Group ≥2 or mean core length ≥4 mm), compared with 15 of 28 cancers (54%) identified by standard biopsy. Targeted biopsies missed clinically significant cancer in two patients compared with five missed clinically significant diagnoses by standard biopsies.

Another prospective study included 347 patients with findings suspicious for prostate cancer, many of whom had one or more previous negative biopsies.[211] All patients received an mpMRI, and those with abnormal findings proceeded to MRI-TRUS fusion-targeted biopsies. The outcome was defined as improved detection in targeted cores, with significantly more cancer detected in targeted cores than in systematic biopsies (30% vs 8.2%). About 12% of those without MRI-suspicious lesions were diagnosed with intermediate-risk tumors. In this study, the cancer detection rate was 51% in individuals with previous negative biopsies.

In a prospective study, 583 patients (56% with prior negative biopsy) underwent mpMRI.[358] All participants received systematic 12-core biopsies, and those with lesions seen on MRI also received fusion-guided biopsies. Multivariate analysis revealed that a higher MRI suspicious score increased the likelihood of finding Grade Group ≥2 cancer by 3.3-fold (95% CI, 2.2–5.1; $P < .0001$).

A meta-analysis of 16 studies (1926 individuals) also showed that MRI-targeted biopsy improved detection of clinically significant prostate cancer

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in those with previous negative biopsies over standard TRUS biopsy.[359] In addition, the use of high-resolution micro-US has been compared to mpMRI and found to perform similarly for the detection of prostate cancer (when performed by experts).[360]

Overall, the Panel believes that targeted biopsy techniques may help identify regions of cancer missed on prior biopsies and should be strongly considered in patients with a prior negative biopsy and persistent concern for cancer based on PSA levels and/or results of biomarker testing.[251]

Saturation Biopsy Techniques

In saturation biopsies, cores are collected systematically every few millimeters across the entire prostate to improve prostate cancer detection over that of a standard 12-core biopsy. In one study, transperineal prostate mapping biopsy was used to calculate biopsy density (the ratio of the total number of specimens retrieved to prostate volume) in 436 patients to determine the optimal sampling approach.[361] Results showed that biopsy density >1.5 was associated with a 1.5-fold higher rate of prostate cancer diagnosis and a higher rate of detection of higher volume Grade Group 1 disease.

Saturation biopsies can be performed via transrectal or transperineal approaches, the latter of which is often image-guided (see *Targeted Biopsy for Repeat Biopsy*, above). The transrectal and transperineal saturation approaches seem to have similar rates of cancer detection.[362] In fact, one study compared the approaches head-to-head and found similar cancer detection rates in the repeat biopsy setting (31.4% for transrectal vs 25.7% for transperineal; $P = .30$).[363] The transperineal approach may have a lower risk of infection, may allow for better saturation of the gland, and may be more acceptable to patients compared with the transrectal approach.[364] In fact, studies reported zero or near-zero rates of sepsis in individuals biopsied with the transperineal approach.[365-367] Another possible benefit of the transperineal over the transrectal approach is more accurate risk assessment (cancer volume and grade).[368] However, the transperineal approach may be associated with a higher rate of urinary retention.[364] The transrectal approach can be performed routinely in the office whereas transperineal biopsy often requires more extensive local or systemic analgesia.

A study of transperineal template-guided mapping biopsy found detection rates of 55.5%, 41.7%, and 34.4% for those with 1, 2, and ≥3 previous negative biopsies, respectively.[369] Other groups have reported similar rates of detection using saturation biopsies in individuals with previous negative biopsies.[367,370,371]

Compared with an extended biopsy approach (12–14 cores), one prospective, non-randomized study found that transrectal saturation biopsy detected significantly more cancers in individuals with one previous negative biopsy (32.7% vs 24.9%; $P = .0075$).[372] The detection of insignificant cancer did not differ significantly between the groups (40.1% vs 32.6%; $P = .20$).

Despite this emerging evidence, the Panel does not recommend a saturation biopsy strategy for all individuals with previous negative biopsies at this time given the benefits seen for MRI and MRI-targeted biopsy in this patient population.

Risks of Biopsy

Biopsies are associated with the risk of complications; one of the least common but potentially most significant clinically is antibiotic-resistant *Escherichia coli* infection.[373] The range of potential infectious complications includes urinary tract infection (UTI), epididymitis, orchitis, prostatitis, and sepsis. In fact, the 30-day risk of invasive *E. coli* infections in U.S. hospitals from 2009 to 2016 was 5.0 per 1000 patients.[374]

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Antimicrobial prophylaxis is therefore generally used with transrectal biopsies to prevent infections and reduce the need for antibiotics after the procedure, thus reducing the overuse of antibiotics.[375] Studies have reported that up to 78% of post-biopsy infections are resistant to fluoroquinolone, many of which are also resistant to other antibiotics.[374,376,377] In addition, the FDA labels for drugs in this class include additional warnings about disabling and potentially permanent side effects of the tendons, muscles, joints, nerves, and central nervous system; risk of ruptures or tears in the aorta blood vessel; serious low blood sugar levels; and mental health side effects.[378] The American Urological Association (AUA) acknowledges that fluoroquinolone prophylaxis should be considered with transrectal biopsy in some centers and under some clinical conditions.[375] Other options recommended by the AUA include first- and second-generation cephalosporin with or without aminoglycoside; and third-generation cephalosporin.[375]

Although these fluoroquinolones-resistant infections will respond to cephalosporins, measures are needed to prevent additional resistant strains. One strategy is to utilize stringent criteria for biopsy as outlined in these guidelines. Another approach is the use of transperineal prostate biopsies, which may be associated with a lower risk of sepsis and a reduced need for antibiotics compared to a transrectal approach; however, data are conflicting[379-381] Results from the prospective, randomized ProBE-PC clinical trial showed no statistical difference in infectious and noninfectious outcomes in individuals who underwent transrectal (n = 351) prostate biopsy versus transperineal (n = 367) prostate biopsy.[382] In contrast, in the randomized TRANSLATE fewer of the 562 participants who received a transperineal biopsy had infections that required hospitalization than of the 564 participants who received a transrectal biopsy (0.4% vs 1.6%).[345] Other proposed strategies include selectively targeted antibiotic prophylaxis with pre-biopsy rectal culture and selectively augmented prophylaxis with two antibiotics in patients at elevated risk for infections.[373,383]

Other morbidities associated with prostate biopsies include rectal bleeding, hematuria, vasovagal episodes, fever, hematospermia, and dysuria.[384,385] Furthermore, up to 90% of individuals undergoing a prostate biopsy have reported some discomfort during the procedure.[386] Both topical lidocaine gel and an injectable nerve block have been shown to be safe and efficacious for reducing discomfort.[387,388] Topical lidocaine was more efficacious in reducing pain during probe insertion, whereas peri-prostatic injection reduced pain during the biopsy itself. Results of one small clinical trial suggest that a combination of lidocaine suppository and periprostatic nerve block might be more effective at reducing pain during prostate biopsy than either one alone.[389] Another small trial found the combination of lidocaine with pelvic plexus block to be most effective at relieving pain associated with prostate biopsy.[390] More recently, a randomized trial compared peri-prostatic nerve block with subcutaneous perineal anesthesia and intrarectal lidocaine gel with total intravenous anesthesia in 216 patients receiving a TRUS-guided transperineal prostate biopsy.[391] The combination treatment resulted in less pain during the biopsy, shorter operation times ($P < .05$), more stable hemodynamics and respiratory status, and fewer surgical complications ($P < .05$).

These minor anesthetic techniques greatly enhance the acceptability of the procedure, particularly with extended templates and saturation techniques, and should be considered in all patients.[392] For situations such as individuals with anal strictures, those who do not readily tolerate biopsy under local anesthesia, or patients who have been inadequately blocked with a periprostatic injection, deep sedation or general anesthesia may be advantageous.

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Management of Biopsy Results

Cancer

Patients diagnosed with prostate cancer by biopsy should receive comprehensive care as per the NCCN Guidelines for Prostate Cancer (available at www.NCCN.org). Among patients diagnosed with cancer on prostate biopsy, the Panel does not recommend routine repeat biopsy, except in special circumstances, such as: 1) there is suspicion that the patient harbors more aggressive cancer than was evident on the initial biopsy; and 2) the patient is otherwise a candidate for active surveillance as outlined in the treatment guidelines.

Intraductal Carcinoma without Invasive Cancer

Intraductal carcinoma (IDC) represents an independent adverse pathologic factor in both radical prostatectomy and needle biopsy specimens that may influence disease response to current therapeutic regimens for advanced-stage prostate cancer. In radical prostatectomy, IDC correlates with other adverse pathologic features in the associated invasive prostate cancer, including higher Gleason score, larger tumor volume, and greater probability of extraprostatic extension, seminal vesicle invasion, and pelvic lymph node metastasis; it also independently predicts biochemical recurrence, progression-free survival, and cancer-specific mortality after radical prostatectomy.[393-401]

In biopsy specimens, IDC is typically seen with high-grade, high-volume prostate cancer and is associated with adverse findings in radical prostatectomy and poor outcomes. IDC diagnosed in prostate biopsies provided an independent prognostication of early biochemical recurrence, cancer-specific survival, survival in patients with distant metastasis at presentation, and the development of metastases after radiation therapy in intermediate- and high-risk prostate carcinoma.[402-404] IDC may be resistant to current therapeutic regimens for aggressive prostate cancer and may require a multimodal approach and novel therapy.[405]

IDC's presence in biopsy material strongly suggests the presence of high-grade cancer. Therefore, proceeding directly to definitive therapy is recommended when IDC is seen on biopsy in the absence of invasive carcinoma (see the NCCN Guidelines for Prostate Cancer, available at www.NCCN.org). Otherwise, careful evaluation is indicated, with strong consideration of a repeat biopsy using MRI targeting to look for invasive cancer.

Atypical Intraductal Proliferation without Invasive Cancer

Intraductal proliferations may show a greater degree of architectural complexity and/or cytologic atypia than typical high-grade prostatic intraepithelial neoplasia (HGPIN; see below), yet they fall short of the strict diagnostic threshold for IDC. The preferred terminology for these lesions is "atypical intraductal proliferation (AIP)."[406-408] When diagnosed on needle biopsy, AIP is potentially considered a marker of unsampled cancer, and it is associated with a 50% increased risk of invasive carcinoma and/or IDC on repeat biopsy.[408,409] One study found that prostate cancer associated with AIP had worse pathologic features than prostate cancer associated with HGPIN.[398] Another study found that AIP-associated prostate carcinoma had similar clinicopathologic features to IDC-associated carcinoma.[407]

When AIP is seen on biopsy in the absence of invasive carcinoma, repeat biopsy using MRI targeting and systematic biopsy to look for invasive carcinoma is recommended.

Other Notable Pathologic Features and Benign Results

Approximately 10% of patients undergoing biopsy will be found to have HGPIN.[410] Cytologically, the nuclear features of HGPIN resemble that of malignant tumors; however, the presence of a basal layer on the acini distinguishes this entity from cancer. Extended biopsy schemes have resulted in a dramatic decline in the prevalence of cancer detected from a repeat biopsy in patients with HGPIN detected from the initial biopsy.

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While reports in the sextant biopsy era demonstrated cancer rates of approximately 50%, contemporary series using extended biopsy schemes report rates of approximately 10% to 20% and occasionally higher.[411-413] Interestingly, the rates of cancer with repeat biopsy in patients with HGPIN seem to differ slightly from those who undergo repeat biopsy based on other risk factors, such as age, family history, and PSA. In addition, most detected cancers are low grade.[414]

Atypia are characterized by glands that either architecturally or cytologically in some ways resemble prostatic adenocarcinoma, but lack full criteria for malignancy. They are distinct from HGPIN because they lack a basal cell layer. Atypia are either diagnosed with descriptive terminology or synonymously designated as atypical small acinar proliferation (ASAP).[415,416] Unlike HGPIN, ASAP is not a distinct pathologic diagnosis; rather it represents either benign mimickers of prostate cancer or cancer that lacks sufficient architectural or cytological atypia to warrant a definitive malignant diagnosis. Approximately 5% of patients receive a biopsy result of atypia. Even in the era of extended biopsy schemes, the prevalence of cancer detected from a repeat biopsy in patients with atypia detected from the initial biopsy is quite high: ≥50%, with the most likely area of cancer detection residing in the prostate area demonstrating atypia from the initial biopsy.[417,418]

Most prostate biopsy results are benign; however, a negative prostate biopsy does not preclude the presence of cancer. Thus, patients with benign results, as well as those with HGPIN or ASAP, may have prostate cancer and require follow-up. However, a balance is needed between maximizing the identification of clinically significant cancer and minimizing over-testing and overdiagnosis. Studies have shown that 4% to 9% of patients with ASAP, HGPIN, or benign biopsy results receive a diagnosis of Grade Group ≥2 prostate cancer on repeat biopsy, with no significant difference between these groups.[413,415,419-421] The rates of Grade Group 1 prostate cancer on repeat biopsy are 24% to 30%.[413,415,420,421] Furthermore, the prostate cancer-specific mortality rate in individuals with initially negative biopsy results is low. In a Danish study, the 15-year prostate cancer-specific mortality rate of patients with an initially negative biopsy was 1.3% for those with PSA <10 ng/mL and 4.6% for those with PSA >20 ng/mL.[422]

For patients with benign biopsy results, atypia suspicious for cancer, or HGPIN, recommendations depend on whether there was a prior high-quality mpMRI. For those with a prior high-quality mpMRI , PSA and DRE are recommended at 12- to 24-month intervals. Certain biomarker tests can also be considered (category 2B; see *Biomarker Testing*, above). For those without prior high-quality mpMRI, mpMRI should be obtained, and biomarker testing can also be considered (category 2B; see *Biomarker Testing*, above). For atypia, repeated biopsy in 12 to 24 months with relative increased sampling of the atypical site should be considered. For all these patients, a repeat prostate biopsy with refined biopsy techniques is recommended if there is a high suspicion of cancer. Characteristics that increase the suspicion of cancer after a negative biopsy are a positive mpMRI, a persistently high PSA, a shorter PSA doubling time, a high PSA density, and/or an abnormal DRE.[423-425]

Summary

Since the early 1990s, many techniques have been introduced in attempts to increase the sensitivity of early detection programs or cancer detection while maintaining specificity (elimination of unnecessary biopsies). These NCCN Guidelines recommend a method by which individuals and their physicians can use these techniques rationally for the early detection of prostate cancer. These guidelines are not designed to provide an argument for the use of population early detection programs for prostate cancer. Rather, they are meant to provide a vehicle by which early detection efforts can be practiced in an evidence-based, systematic

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fashion in patients who choose to participate in such programs. Whether to treat a patient upon diagnosis is beyond the scope of these guidelines (see the NCCN Guidelines for Prostate Cancer, available at www.NCCN.org).

These NCCN Guidelines for Prostate Cancer Early Detection will incorporate recently validated findings when they occur. The Panel will re-examine the clinical utility of new modalities annually, and the guidelines will be modified accordingly.

The goal of NCCN and this Guidelines Panel in updating these algorithms is to assist patients and clinicians in choosing a program of early detection for prostate cancer and in making decisions regarding the need for prostate biopsy. Any clinician who uses these guidelines is expected to exercise independent medical judgment in the context of the individual clinical circumstances to determine the patient's need for prostate biopsy. These guidelines will continue to evolve as the field of prostate cancer advances.

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